



Novel progressive failure model for quasi-orthotropic pultruded FRP structures: Formulation and calibration of parameters (Part I)

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ABSTRACT

This paper presents a novel progressive failure model for the 3D simulations of pultruded FRP structures which allows the modelling of the laminates as a homogeneous material. The failure initiation model proposed requires only the strength in each direction as input, combining them to retrieve in-plane and out-of-plane failure indexes. The damage propagation model can be divided in two main stages: (i) damage progression and (ii) constant stress beyond a limit strain. The former stage uses the in-plane and out-of-plane failure indexes to determine the damage progression, using different parameters in each direction to account for the different damage responses, while the latter is characterized by a constant stress after a limit strain is reached, also different for each direction. FE models were developed with the proposed damage propagation model, requiring as input the strengths obtained from standardized experimental material coupon testing, the results of which, namely the load/stress vs. displacement/strain curves, are used to calibrate all the parameters needed to establish the model. The results show that the proposed damage propagation model, using a homogenized material, is well able to predict the experimental behaviour even for very complex cases such as interlaminar shear tests. Furthermore, in a companion paper the accuracy and limitations of the model are further assessed in the simulation of transverse compact tension and web-crippling tests.

1. Introduction

During the last half of the 20th century, the structural use of fibre reinforced polymer (FRP) composites significantly increased across a wide range of industries, from aerospace to construction. In order to correctly design and efficiently use these structural materials, simulation models have been developed to predict their failure behaviour, initially by adapting existing models, developed for isotropic materials, to account for the apparent orthotropy of FRPs [1,2].

In 1950, Hill [3] presented one of the earlier failure models for fully anisotropic materials, which however presented the difficulty of requiring several parameters that are difficult to obtain experimentally. In the next decade, Tsai [4] proposed an adaptation of Hill's model in order to include the different tensile and compressive strengths of FRP materials, known as the Tsai-Hill model, which is valid for orthotropic materials. Later, Tsai and Wu [5] proposed a more generalized failure model, which, unlike the Tsai-Hill model, is valid for anisotropic materials, and not only for orthotropic materials.

On the other hand, the application of this model to non-transversely isotropic materials (e.g. FRPs with a significant content of fibres in non-longitudinal directions) requires the experimental determination of equibiaxial strengths, which has proven to be very difficult and onerous, as attested by the very little experimental results available [6]. In 1980, Hashin [7] proposed a set of failure criteria for unidirectional composites (*i.e.* transversely isotropic), separating fibre (longitudinal) and matrix (transverse) failure in both tension and compression. The criteria were developed based on the invariants of the stress tensor (assuming that the longitudinal direction was always a main direction), which were then combined quadratically. Many other failure models for FRP materials were proposed [1,2] and, in the last decades, more accurate, yet much more complex, failure initiation models have also been developed, such as LaRC03 [8] or Puck [9]. In order to correctly use these models in multi-layered laminates, ply discretization is required, with high computational costs and, as pointed-out by Talreja [10], requiring the experimental determination of a large set of parameters. Moreover, owing to their relative simplicity, the Tsai-Hill [4],

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Tsai-Wu [5] and Hashin [7] failure initiation models have been the most widely used [1,11], including in the simulation of non-transversely isotropic materials, such as pultruded FRPs, thus outside their limit of validity, or, in the case of the Tsai-Wu model, without considering all the experimental parameters needed as input. Moreover, the application of these models to pultruded FRP members has showed that they tend to underestimate the experimental strength (e.g. [12–15]), highlighting the need to take into account damage propagation in the failure analysis.

Matzenmiller et al. [16] proposed an exponential damage evolution model (often referred to as MLT), coupled to a failure initiation criterion, which has been proven to present fairly accurate results [17], requiring the parameterization of damage coefficients that define the “smoothness” of the damage evolution. Another approach was pursued by Camanho and Dávila [18], who developed a decohesion finite element (FE) in order to apply a mix-mode progressive damage model to simulate delamination phenomena between solid elements. The simulation of progressive damage developed by these authors entailed a progressive strain-softening model, which was defined by the fracture energy, a concept successfully used in concrete [19]. This approach, in comparison to that proposed by Matzenmiller et al. [16], has the advantage of using parameters with physical significance, such as fracture energies. On the other hand, the experimental determination of fracture energies, one for each failure mode, may be very difficult, requiring specific (often non-standardized) characterization tests. Moreover, unlike what occurs in concrete, the energy values obtained may present a very wide range for comparable materials [20–22].

The aforementioned models have been widely used in numerical analysis and some were included in commercial software packages. ABAQUS [23], in particular, has a built-in damage initiation and progression model, developed by Lapczyk and Hurtado [24], which is based on the works of Hashin [7], Matzenmiller et al. [16] and Camanho and Dávila [18]. However, owing to its transverse isotropic assumption, this model is only available for elements with shell formulation, greatly limiting its application to complex structures, involving contacts, such as connections (e.g. [25]), and its use with non-transversely isotropic composites (although possible) is theoretically not valid. Additionally, as mentioned, in order to capture through-thickness damage, such as delamination, ply modelling is required with prescription of cohesive failure laws between plies, thus increasing computational costs and requiring the input of extra damage parameters.

Moreover, recent research (e.g. [26–29]), based on the aforementioned works, has been focused on the employment or development of damage progression models for finite element (FE) analysis, which are in fact able to predict the failure of FRP laminates and, in some cases, their post-failure behaviour. However, these models often rely on the modelling of the individual plies of the laminates and/or in the prescription of cohesive planes (defined *a priori*), resulting in high density meshes and high computational costs, rendering them impractical for the analysis of full-scale engineering problems. Additionally, most studies have focused on quasi-isotropic thin laminates (<5 mm), which means they may not be applicable for most pultruded FRP products, namely those used in civil infrastructure (wall thicknesses typically higher than 8–10 mm).

In this context, this paper presents a novel damage progression model for the 3D FE analysis of homogenised FRP laminates and, in particular, of pultruded quasi-orthotropic FRP elements. Since these elements are available off-the-shelf (i.e. as a final product) and their exact constitution (fibre architecture, resin matrix, etc.) is not disclosed by manufacturers, layered modelling is in practice not possible. This justified the development of a new 3D failure initiation model to evaluate the in-plane and out-of-plane failure of FRP laminates taken as homogenous (through the thickness). In this model, the failure progression is simulated by (i) reducing the initial elastic and shear moduli, following an adapted MLT model, which considers the different directions of damage and the stress state (compression or tension),

and (ii) setting a residual constant stress beyond a limit strain. The calibration of the parameters required by the model can be obtained from experimental results of standard material characterization tests, which must always be performed to determine the elastic and strength properties of composite materials. The first part of this two-part paper presents and discusses the proposed damage progression model and its calibration parameters for two laminates corresponding to the web and flanges of an I-shaped pultruded glass fibre reinforced polymer (GFRP) profile. The second part of the paper [30] presents the application of the model to simulate two different problems: (i) compact tension tests, and (ii) web-crippling tests.

2. Proposed model: Formulation

2.1. Failure initiation model

2.1.1. Description of the model

The failure initiation model proposed herein is divided in two criteria for (i) in-plane and (ii) out-of-plane failure. In both cases, the relevant stresses are combined in quadratic form, similarly to previously proposed failure models (e.g. [4,7]). In order to account for the different strength of pultruded FRP materials in compression and tension (for all orthogonal directions), ellipsoid envelopes are proposed; thereafter, the envelope for in-plane failure is given by Eq. (1),

$$F_{pl}(\sigma) = \left(\frac{\sigma_{11} - C_{11}}{A_{11}} \right)^2 + \left(\frac{\sigma_{22} - C_{22}}{A_{22}} \right)^2 + \left(\frac{\tau_{12}}{A_{12}} \right)^2 = 1 \quad (1)$$

and the envelope for out-of-plane failure is given by Eq. (2),

$$F_{op}(\sigma) = \left(\frac{\sigma_{33} - C_{33}}{A_{33}} \right)^2 + \left(\frac{\tau_{13}}{A_{13}} \right)^2 + \left(\frac{\tau_{23}}{A_{23}} \right)^2 = 1 \quad (2)$$

where σ_{ii} and τ_{ij} represent the normal and shear stresses in directions ii and ij , respectively. In this paper, the longitudinal direction is represented by 1, the in-plane transverse direction by 2 and the out-of-plane direction by 3. C_{ii} stand for the coordinate of the envelopes' centres in the ii direction, given by,

$$C_{ii} = \frac{S_{ii}^+ - |S_{ii}^-|}{2} \quad (3)$$

where S_{ii}^+ and S_{ii}^- represent the tensile and compressive strengths in direction i , respectively. Additionally, A_{ij} is the maximum distance between the centre of the ellipsoid and the failure surface over a given ij axis, and may be determined by the following expressions,

$$A_{11} = \frac{S_{11}^+ - C_{11}}{\sqrt{1 - \left(\frac{C_{22}}{A_{22}} \right)^2}} \quad (4)$$

$$A_{22} = \frac{\sqrt{(S_{22}^+ - C_{22})^2 - \left(\frac{C_{11}C_{22}}{S_{11}^+ - C_{11}} \right)^2}}{\sqrt{1 - \left(\frac{C_{11}}{S_{11}^+ - C_{11}} \right)^2}} \quad (5)$$

$$A_{12} = \frac{S_{12}}{\sqrt{1 - \left(\frac{C_{11}}{A_{11}} \right)^2 - \left(\frac{C_{22}}{A_{22}} \right)^2}} \quad (6)$$

$$A_{33} = S_{33}^+ + |S_{33}^-| \quad (7)$$

$$A_{13} = \frac{S_{13}}{\sqrt{1 - \left(\frac{C_{33}}{A_{33}} \right)^2}} \quad (8)$$

$$A_{23} = \frac{S_{23}}{\sqrt{1 - \left(\frac{C_{33}}{A_{33}} \right)^2}} \quad (9)$$

where S_{ij} is the shear strength in the plane ij . The failure envelopes are depicted in Fig. 1 for both in-plane and out-of-plane failure. This approach allows for the definition of a continuum function with continuum derivative for in-plane and out-of-plane failure, which is not the case of some widespread failure models, such as Hashin failure criteria [7,31].

Thereafter, failure indexes for the in-plane (f_{pl}) and out-of-plane (f_{op}) failure can be derived from Eqs. (10a) and (10b), respectively.

$$F_{pl}\left(\frac{\sigma}{f_{pl}}\right) = 1 \quad (10a)$$

and

$$F_{op}\left(\frac{\sigma}{f_{op}}\right) = 1 \quad (10b)$$

The solution of Eqs. (10a) and (10b) renders,

$$f_{pl}(\sigma) = \frac{-b_{pl} - \sqrt{b_{pl}^2 - 4a_{pl}c_{pl}}}{2a_{pl}} \quad (11a)$$

and

$$f_{op}(\sigma) = \frac{-b_{op} - \sqrt{b_{op}^2 - 4a_{op}c_{op}}}{2a_{op}} \quad (11b)$$

and

$$a_{pl} = \left(\frac{C_{11}}{A_{11}}\right)^2 + \left(\frac{C_{22}}{A_{22}}\right)^2 - 1 \quad (12)$$

$$b_{pl} = -2\left(\frac{C_{11}\sigma_{11}}{A_{11}^2} + \frac{C_{22}\sigma_{22}}{A_{22}^2}\right) \quad (13)$$

$$c_{pl} = \left(\frac{\sigma_{11}}{A_{11}}\right)^2 + \left(\frac{\sigma_{22}}{A_{22}}\right)^2 + \left(\frac{\tau_{12}}{A_{12}}\right)^2 \quad (14)$$

$$a_{op} = \left(\frac{C_{11}}{A_{11}}\right)^2 - 1 \quad (15)$$

$$b_{op} = -2\frac{C_{33}\sigma_{33}}{A_{33}^2} \quad (16)$$

$$c_{op} = \left(\frac{\sigma_{33}}{A_{33}}\right)^2 + \left(\frac{\tau_{13}}{A_{13}}\right)^2 + \left(\frac{\tau_{23}}{A_{23}}\right)^2 \quad (17)$$

It should be mentioned that the mechanical parameters needed to establish this model, *i.e.* the various strengths, can be determined with standardized (relatively simple) coupon tests, except for the through-thickness tensile and compressive strengths, for which no standardized tests are yet available.

2.1.2. Discussion

The present model allows for the straightforward definition of two failure envelopes, for in-plane and out-of-plane failure, based on the strength on each direction, encompassing the normal compressive and tensile stresses in the same expressions. Besides the simplicity of its definition, the envelopes proposed have continuum derivative throughout their domain, unlike most models been proposed in recent years [31]. On the other hand, it should be stressed that the proposed formulation effectively uncouples in-plane and out-of-plane failure. This is a reasonable simplification of the very complex failure mechanisms of FRP materials for laminates presenting quasi-orthotropic behaviour, which includes most pultruded materials that generally present a very large fibre content in the longitudinal direction. Moreover, due to their aspect ratio, the laminates of commercial pultruded profiles can be considered as thin plates and, thus, in-plane stresses are typically much higher than their out-of-plane counterparts.

The present model is well able to account for the effect of normal stresses in shear strength, which has been one of the main focus of development in failure initiation models. With this respect, Dávila et al. [8] present a comparison between the predictions of several models with experimental results reported by Soden et al. [32] for the normal and shear stress interaction on the transverse (1–2) plane. Fig. 2 presents the same comparison, including the predictions obtained with the model proposed herein. These results show that the present model has a good agreement with the experimental scatter. It should be mentioned that more complex models, such as LaRC03 [8] and Puck [9], present a better agreement with the experimental data when the compression axial load is around 50% of the maximum compressive strength. However, in order to do so, these models require as input the fracture angle and the fracture energies, which must be obtained from additional (and complex) experimental testing, whose results are generally not readily available. Furthermore, as mentioned, unlike the present model, none of the models presented in Fig. 2 has a continued derivative throughout their domain.

Overall, the present model presents a very good balance between accuracy and limitations and its ease of implementation, requiring only the individual strengths as input, thus making it as a valuable potential tool for the design of FRP composites.

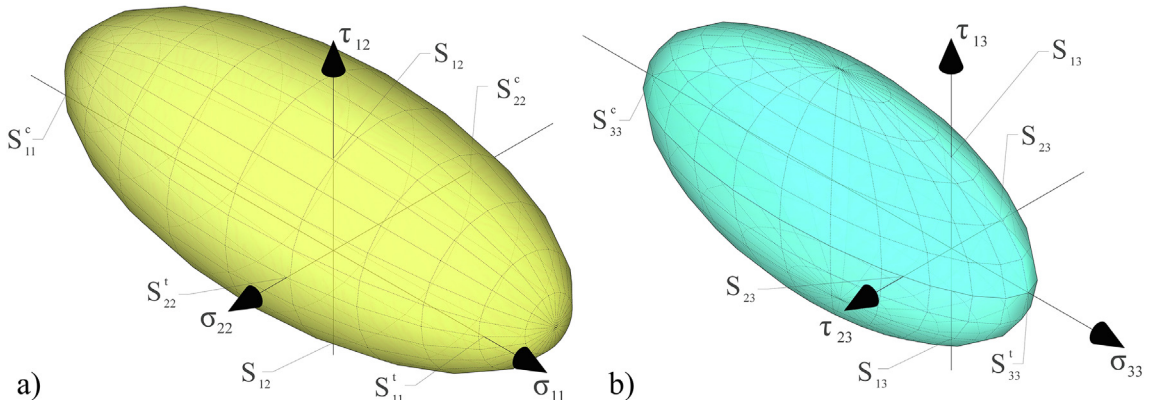


Fig. 1. Failure envelopes: a) in-plane failure and; b) out-of-plane failure.

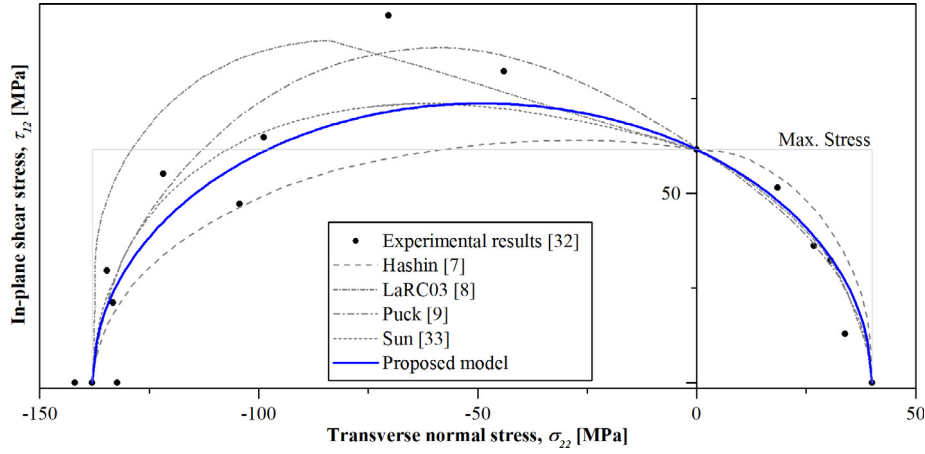


Fig. 2. Comparison between failure envelopes and experimental data. adapted from [8]. See above-mentioned references for further information.

2.2. Damage propagation model

2.2.1. Description of the model

To evaluate the failure behaviour of an FRP structure, namely to predict its strength, the use of failure initiation models generally leads to inaccurate estimates. In fact, localized failure modes can occur (involving significant stress redistribution) without compromising the overall structural integrity. Thereafter, in order to evaluate the behaviour of these structures up to failure and their responses after damage initiation, damage propagation models must be employed.

FRP materials are often described as brittle or quasi-brittle since they often exhibit an almost linear behaviour up to failure and low strength beyond a limit strain (ϵ_u), particularly in tension. On the other hand, these materials have demonstrated ability to present progressive failure modes, namely in compression (crashworthiness, e.g. [34,35]).

In this study, in order to account for the different possible stages of response of FRP materials, the stress–strain relationship was idealized as having three stages; this is shown in Fig. 3, for a one directional illustrative case. The first stage, before the damage on-set, is character-

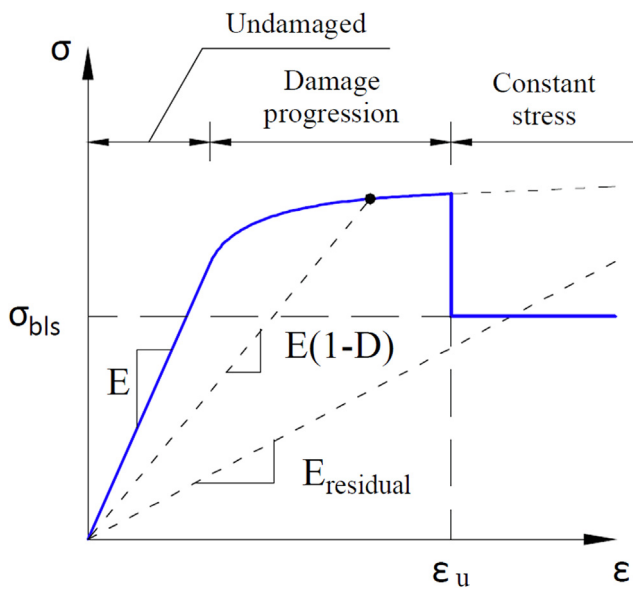


Fig. 3. Idealized stress–strain relationship (for one directional case) with three stages: (i) undamaged; (ii) damage progression, and (iii) constant stress (σ_{bls}) beyond a limit strain.

ized by linear behaviour, according to the (initial) elastic modulus (E). After damage is initiated, the elastic modulus starts to decrease, as a function of a damage variable (D), tending to a residual elastic modulus ($E_{residual}$), which corresponds to the maximum damage allowed at this stage (d_{max}). Finally, after a certain limit strain (ϵ_u) is reached, the behaviour is characterized by a constant (residual) strength (σ_{bls}).

In order to implement the aforementioned damage progression model, and in particular for the damage progression stage (cf. Fig. 3), the authors used an exponential damage law proposed by Matzenmiller et al. [16], often referred to as the Matzenmiller-Lubliner-Taylor (MLT) model. However, in the present work, recognizing the different damage and failure responses in each direction, as well as their importance to the global behaviour, a specific damage value is attributed to each moduli (i.e. E_{11} , E_{22} , E_{33} , G_{12} , G_{13} and G_{23}). Moreover, for the Young's moduli, different damage progression and maximum damage can be attributed for compression and tension. Finally, the damage evolution variables can be written as a function of the failure indexes (cf. Eq. (11)), for the in-plane and out-of-plane directions, respectively,

$$D_i = d_{i,max}^{\pm} \left(1 - e^{-\frac{f_{pl}(\hat{\sigma})^{m_i^{\pm}}}{m_i^{\pm} e}} \right) \quad i = 1, 2, 4 \quad (18)$$

$$D_i = d_{i,max}^{\pm} \left(1 - e^{-\frac{f_{op}(\hat{\sigma})^{m_i^{\pm}}}{m_i^{\pm} e}} \right) \quad i = 3, 5, 6 \quad (19)$$

where $d_{i,max}^{\pm}$ and $m_i^{\pm}$ are the maximum damage allowed at this stage and the exponential damage evolution control variable, respectively, for the i direction, and the superscript $\pm$ indicates the direction of the load (+ for tension and – for compression), relevant for directions 1 to 3 only. Additionally, it should be mentioned that the failure indexes in Eqs (18) and (19) are computed for the effective stresses ($\hat{\sigma}$), which are discussed ahead in this section.

Fig. 4 shows the evolution of a generic damage variable (D) with the exponential damage evolution control variable (m), showing that for lower m values, the damage onset occurs for low failure indexes (f), effectively reducing the extension of the “undamaged” stage (cf. Fig. 3). This effect can be helpful when the stress–strain relationship presents non-linear behaviour since early stages (e.g. transverse axial response of quasi-unidirectional FRPs). As the damage evolution control variable grows, the damage onset tends to failure indexes of 1. Conversely, after that value is reached, large m values increase the damage variable abruptly, rapidly reaching the maximum damage

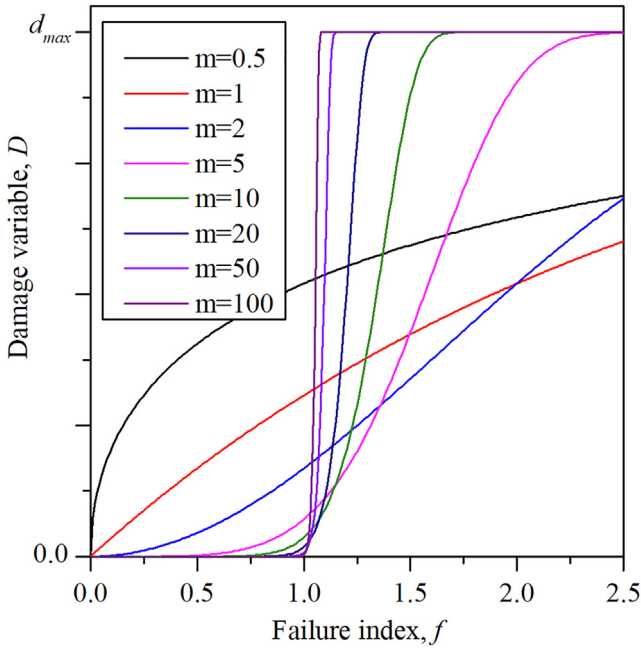


Fig. 4. Generic evolution of the MLT damage variable (D) against a failure index (f).

(d_{max}), while lower m values show a smother progression, reaching the maximum damage for failure indexes considerably larger than 1.

In the proposed model, Eqs. (18) and (19) are valid for the damage progression stage, which, as mentioned earlier, ends when a limit strain ($\varepsilon_{ii,u}^{\pm}$ or $\gamma_{jk,u}$) is reached, after which the stress-strain relationship is governed by a constant stress (cf. Fig. 3). At that stage, the damage variables verify one of the following conditions,

$$D_i = 1 - \frac{r_i^{\pm} S_{ii}^{\pm}}{E_{ii} \varepsilon_{ii}} \quad i = 1, 2, 3 \quad (20)$$

$$D_i = 1 - \frac{r_i^{\pm} S_{jk}^{\pm}}{G_{jk} \gamma_{jk}} \quad i = 4, 5, 6 \quad jk = 12, 13, 23 \quad (21)$$

where $r_i^{\pm}$ is the strength retention percentage relative to each damage variable and the superscript $\pm$ indicates the direction of the load, relevant for i equal to 1 to 3, E_{ii} and G_{jk} are the elastic and shear moduli, and ε_{ii} and γ_{jk} are the total absolute strains for directions ii and jk , respectively.

As can be retrieved from Eqs. (20) and (21), once the strains in one of the directions (or planes for the shear strains) reaches the limit set, the damage variables of that direction are not directly correlated with those of the remaining directions, unlike what occurs in the damage progression stage. This is a simplification of this very complex problem, justified by the high damage level of the FRP material at this stage, allowing for a simpler definition of its post-failure behaviour. Nevertheless, the effect of damage in the normal directions is still indirectly correlated with the damaged secant stiffness tensor (C_d), which is defined by Eq. (22),

$$C_d = M \cdot C \quad (22)$$

where C represents the undamaged secant stiffness tensor,

$$C = \begin{bmatrix} \frac{1-v_{23}v_{32}}{E_{22}E_{33}\Delta} & \frac{v_{21}+v_{31}v_{23}}{E_{22}E_{33}\Delta} & \frac{v_{31}+v_{21}v_{23}}{E_{22}E_{33}\Delta} & 0 & 0 & 0 \\ \frac{v_{12}+v_{13}v_{32}}{E_{11}E_{33}\Delta} & \frac{1-v_{31}v_{13}}{E_{11}E_{33}\Delta} & \frac{v_{13}+v_{12}v_{32}}{E_{11}E_{33}\Delta} & 0 & 0 & 0 \\ \frac{v_{13}+v_{12}v_{23}}{E_{11}E_{22}\Delta} & \frac{v_{23}+v_{13}v_{21}}{E_{11}E_{22}\Delta} & \frac{1-v_{12}v_{21}}{E_{11}E_{22}\Delta} & 0 & 0 & 0 \\ 0 & 0 & 0 & G_{12} & 0 & 0 \\ 0 & 0 & 0 & 0 & G_{13} & 0 \\ 0 & 0 & 0 & 0 & 0 & G_{23} \end{bmatrix} \quad (23)$$

where v_{jk} represents the Poisson's coefficient between directions j and k , and Δ is given by,

$$\Delta = \frac{1 - v_{12}v_{21} - v_{13}v_{31} - v_{23}v_{32} - 2 \cdot v_{21}v_{32}v_{13}}{E_{11}(1-D_1)E_{22}(1-D_2)E_{33}(1-D_3)} \quad (24)$$

and the effective damage tensor (M) in Eq. (22) can be represented as,

$$M = \begin{bmatrix} 1-D_1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1-D_2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1-D_3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1-D_4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1-D_5 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1-D_6 \end{bmatrix} \quad (25)$$

As mentioned, it should be stressed that the failure indexes used to determine the damage evolution variables (cf. Eqs. (18) and (19)) are computed as a function of the effective stresses ($\hat{\sigma}$), which are obtained by multiplying the inverse of the effective damage tensor by the nominal stresses (σ), according to Eq. (26),

$$\hat{\sigma} = M^{-1} \cdot \sigma \quad (26)$$

where the nominal stresses can be expressed as

$$\sigma^T = \{ \sigma_{11} \quad \sigma_{22} \quad \sigma_{33} \quad \tau_{12} \quad \tau_{13} \quad \tau_{23} \} \quad (27)$$

2.2.2. Discussion

The damage evolution model proposed sets two basic stages, (i) damage evolution and (ii) constant stress beyond a limit strain, aiming at representing the complex failure of composite materials considering a homogeneous material. It should be mentioned that, in the present formulation, the initial “undamaged” stage represented in the idealized stress-strain relationship presented in Fig. 3 does not actually exist, since the MLT degradation model is employed until the maximum strain is reached. However, except for very low exponential damage evolution control variable ($m_i^{\pm}$) values (cf. Fig. 4), in the initial stages of the stress-strain relationship the damage is negligible.

Using the failure initiation model proposed earlier (Section 2.1), in-plane and out-of-plane failure are considered, while the different effects on each direction can be duly taken in account by adjusting the parameters of the MLT and strength beyond a limit strain. In this damage evolution model, (i) the in-plane damage and out-of-plane damage are uncoupled for the damage evolution stage, while in (ii) the constant stress beyond a strain limit stage damage is uncoupled for all directions. The former simplification is reasonable when applying the model to quasi-orthotropic materials with slender aspect ratios, such as most pultruded FRP profiles, as discussed earlier (cf. Section 2.1.2). The latter simplification assumes that, beyond a given strain limit in one direction, (large) damage tends to concentrate in that direction and damage coupling is negligible; on the other hand, the axial damages (D_1 , D_2 and D_3) have an indirect “coupling” as they affect the non-diagonal elements of the damaged secant stiffness tensor (C_d).

Regarding the damaged secant stiffness tensor (C_d), it should be mentioned that the present formulation may lead to non-symmetric tensors, owing to the adopted diagonal effective damage tensor (M). From a computational point, this is not an ideal solution, however, it should be stressed that preliminary tests showed that the adoption of a non-symmetric tensor had little effect on the computation times required for the analyses, namely considering non-uniform stress state cases. On the other hand, the adoption of a symmetric damaged secant stiffness tensor led to convergence difficulties in the transition to the post-peak behaviour when testing uniform stress state cases.

Overall, the present formulation has the advantage of allowing the definition of the damage parameters based solely on the results of the material characterization coupon tests, which are always required for

design, without the need of additional tests, such as those required to define different fracture energies.

3. FE implementation

In order to correctly use the damage propagation model proposed herein, it should be acknowledged that FE analysis often entails mesh sensitivity issues, which occur (regardless of using damage models) in situations where strong discontinuities may exist [36], such as “hard boundaries”, contacts, corners or cracks. Thereafter, to tackle this issue, the transition between the damage progression and constant strength stages, which in the proposed model occurs abruptly (cf. Fig. 3), may be smoothened as a function of the mesh size, similarly to the mesh regularization scheme proposed by Bazant [19]. Fig. 5 illustrates, for a one directional case, the proposed damage model with the added regularization scheme. In order to define this stage, a regularization strain ($\epsilon_{ii,u,eq}^{\pm}$ or $\gamma_{jk,u,eq}^{\pm}$), for which the strength is zero, must be defined,

$$\epsilon_{ii,u,eq}^{\pm} = \epsilon_{ii,u}^{\pm} \times \alpha / L_c \quad \text{or} \quad \gamma_{jk,u,eq}^{\pm} = \gamma_{jk,u}^{\pm} \times \alpha / L_c \quad (28)$$

where α is a multiplying factor that may be calibrated with experimental data, and L_c is the characteristic length of each finite element. The interception between the segments connecting the point of maximum stress to that of the regularization strain and the constant stress, may be found from,

$$\epsilon_{ii,u,eq}^{\pm} = \epsilon_{ii,u}^{\pm} \left[1 + \left(1 - \frac{r_i^{\pm} S_{ii}^{\pm}}{\sigma_{ii}^{\max}} \right) (\alpha / L_c - 1) \right] \quad i = 1, 2, 3 \quad (29)$$

$$\gamma_{jk,u,eq}^{\pm} = \gamma_{jk,u}^{\pm} \left[1 + \left(1 - \frac{r_i^{\pm} S_{jk}^{\pm}}{\sigma_{jk}^{\max}} \right) (\alpha / L_c - 1) \right] \quad i = 4, 5, 6 \quad jk = 12, 13, 23 \quad (30)$$

where $\sigma_{ii}^{\max}$ and $\sigma_{jk}^{\max}$ are the stresses when the input limit strains, $\epsilon_{ii,u}^{\pm}$ and $\gamma_{jk,u}^{\pm}$, respectively, are attained; these can be readily estimated assuming a one-directional stress state. Finally, the damage variables between $\epsilon_{ii,u}^{\pm}$ and $\epsilon_{ii,u,eq}^{\pm}$ (or $\gamma_{jk,u}^{\pm}$ and $\gamma_{jk,u,eq}^{\pm}$) are given by,

$$D_i = 1 - \left[\frac{\sigma_{ii}^{\max} - \left(\frac{\sigma_{ii}^{\max} - r_i^{\pm} S_{ii}^{\pm}}{\epsilon_{ii,u,eq}^{\pm} - \epsilon_{ii,u}^{\pm}} \right) (\epsilon_{ii,u}^{\pm} - \epsilon_{ii,u,eq}^{\pm})}{\epsilon_{ii,u,eq}^{\pm} - \epsilon_{ii,u}^{\pm}} \right] (\epsilon_{ii,u}^{\pm})^{-1} \quad i = 1, 2, 3 \quad (31)$$

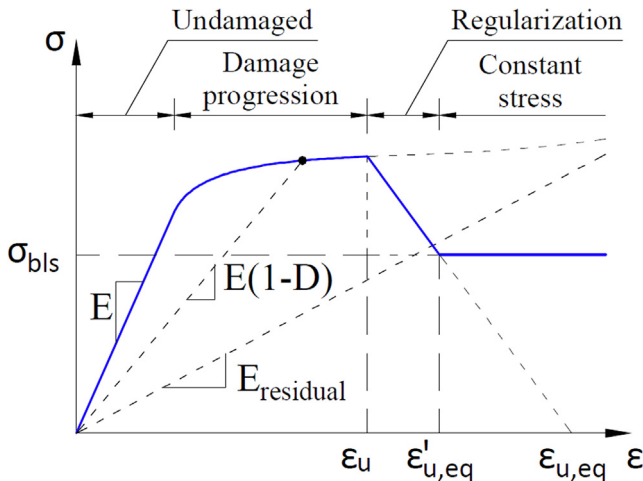


Fig. 5. Implemented stress-strain relationship (for one directional case) with four stages: (i) undamaged; (ii) damaged progression; (iii) mesh regularization; and; (iv) constant stress (σ_{bls}) beyond a limit strain.

$$D_i = 1 - \left[\frac{\tau_{jk}^{\max} - \left(\frac{\tau_{jk}^{\max} - r_i^{\pm} S_{jk}^{\pm}}{\gamma_{jk,u,eq}^{\pm} - \gamma_{jk,u}^{\pm}} \right) (\gamma_{jk,u}^{\pm} - \gamma_{jk,u,eq}^{\pm})}{\gamma_{jk,u,eq}^{\pm} - \gamma_{jk,u}^{\pm}} \right] (\gamma_{jk,u}^{\pm})^{-1} \quad (32)$$

$$i = 4, 5, 6 \quad jk = 12, 13, 23$$

Moreover, in order to avoid severe convergence difficulties in implicit analyses, and following the ABAQUS documentation recommendations [23], a viscous regularization scheme was implemented to ensure damage stabilization where needed, such that the final damage respects the following condition,

$$D_i = \frac{\eta_i^{\pm}}{\eta_i^{\pm} + \Delta t} \cdot D_{i,PS} + \frac{\Delta t}{\eta_i^{\pm} + \Delta t} \cdot D_i \quad (33)$$

where $\eta_i^{\pm}$ is the i^{th} viscous regularization coefficient and the $\pm$ superscript indicates the direction of the load, relevant for i equal to 1 to 3. The time increment is represented by Δt , while $D_{i,PS}^{\pm}$ is the damage obtained in the previous step.

Overall, the model requires a total of 64 input values, including (i) 9 elasticity inputs (elastic and distortion moduli and Poisson's coefficients); (ii) 9 strength inputs; (iii) 18 damage progression control inputs ($d_{i,max}^{\pm}$ and $m_i^{\pm}$); (iv) 18 residual strength control inputs ($\epsilon_{ii,u}^{\pm}$, $\gamma_{jk,u}^{\pm}$ and $r_i^{\pm}$); (v) 1 mesh regularization parameter (α), which is independent of the direction and, additionally; (vi) 9 viscous regularization coefficients.

It should be acknowledged that the input list of the proposed model is extensive, in particular the 36 input variables directly related to the damage progression and residual strength. However, as shown in the following section, those input values may be determined by curve-fitting the experimental results of standard coupon material characterization tests, which are not only easy to carry out but are generally required for any analysis or design.

4. Calibration of parameters

4.1. Experimental programme

To assess the ability of the proposed model to simulate the failure behaviour and damage progression of FRP composites, in particular of pultruded GFRP composites, a set of standardized coupon tests was performed on two different GFRP plates in order to retrieve the experimental data needed to calibrate the model.

The pultruded plates used in the experimental tests were made of E-glass fibres and an isophthalic polyester resin matrix, produced by ALTO, Perfis Pultrudidos, Lda. These plates were obtained from the flanges and web of a $150 \times 75(8) \text{ mm}^2$ I-shaped pultruded profile (I150-F and I150-W, respectively), which was ordered to have the same fibre architecture on the web and flanges. The calcination of the plates up to 800°C allowed to identify uni-directional rovings (R), 0/90 woven (W) and chopped fibre mats (C), with a C/W/R/C/R/W/C layout. The fibre weight proportion per direction was estimated in $\sim 78\%$ for 0° direction, $\sim 4\%$ for 90° direction and $\sim 17\%$ for the chopped fibre mats, confirming the quasi-orthotropy of the laminates.

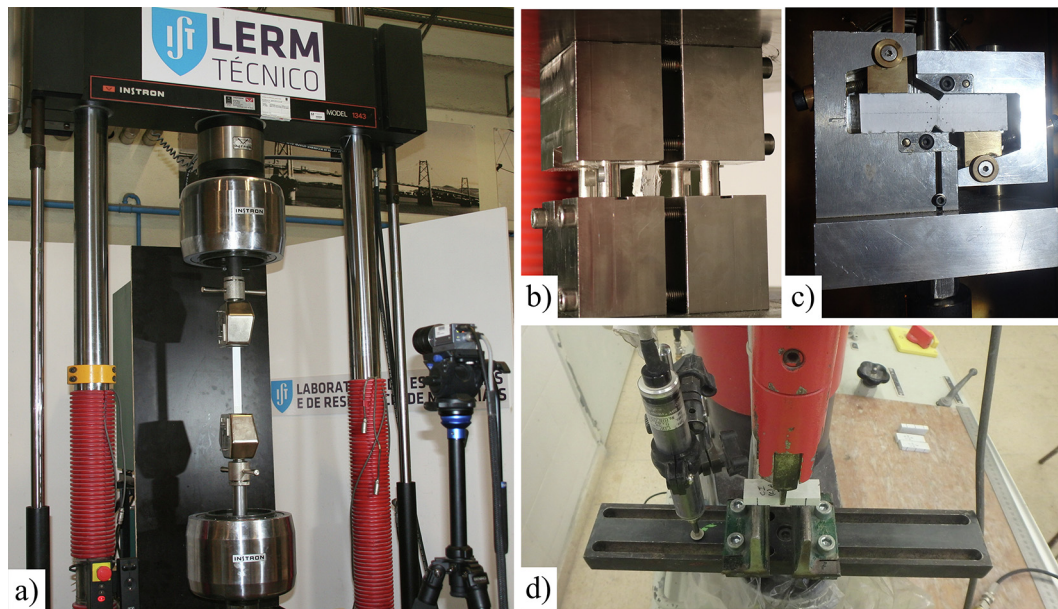
Table 1 lists the tests performed on each plate, summarizing the main results, while Fig. 6 presents an overview of each test. It should be noted that, owing to their reduced dimensions, in particular their low width, it was not possible to perform all tests in all directions and, in some cases, identified in Table 1, it was not possible to fulfil the standard specimen dimensions. Additionally, it should be mentioned that the transverse compression tests were performed with the combined loading compression (CLC) apparatus; however, due to the short length of the specimens, load was only transferred through friction. Due to doubts about the results of these tests, the authors decided to perform additional compressive tests in the transverse direction, as explained in Section 4.3.4. Specific details of each test

Table 1

List of coupon material characterization tests and their main results (average and co-variation).

Test	Standard	Property	Unit	I150-W			I150-F		
				No. Specimens	Average	CoV	No. Specimens	Average	CoV
Longitudinal tension	EN ISO 527 [37]	S_{11}^+	MPa	6	387.97	6%	4	353.47	9%
		E_{11}	GPa		43.36	2%		39.59	3%
		ν_{12}	–		0.23	8%		0.29	5%
Longitudinal compression	ASTM-D6641 [38]	S_{11}^-	MPa	5	461.94	7%	0	–	–
		E_{11}	GPa		44.87	4%		–	–
		ν_{12}	–		0.30	24%		–	–
Transverse tension	EN ISO 527 ¹ [37]	S_{22}^+	MPa	3	44.66	4%	0	–	–
		E_{22}	GPa		9.39	12%		–	–
		$S_{22,y}^+$	MPa		15.18	13%		–	–
Transverse compression ²	ASTM-D6641 ³ [38]	S_{22}^-	MPa	4	64.18	3%	0	–	–
		E_{22}	GPa		8.07	8%		–	–
In-plane shear	ASTM-D5379 [39]	S_{12}	MPa	5	46.76	7%	5	47.89	6%
		G_{12}	GPa		3.01	8%		3.74	8%
Longitudinal interlaminar shear	ASTM-D2344 [40]	S_{13}	MPa	5	27.01	5%	5	31.22	3%
		K_{13}	MPa/mm		65.89	3%		68.66	7%
Transverse interlaminar shear		S_{33}	MPa	3	8.09	8%	4	6.34	19%
		K_{23}	MPa/mm		26.72	7%		24.98	20%

Dimensional restrictions:

¹ These specimens were shorter than recommended by the standard, with 122 mm and 82 mm of total and free lengths, respectively.² Additional compressive tests were performed in the transverse direction following ASTM D 695 [41] (cf. Section 4.3.4).³ These specimens were shorter than recommended by the standard, with 123 mm and 16 mm of total and free lengths, respectively.**Fig. 6.** Experimental setup for: a) tensile; b) combined loading compression (CLC); c) V-notched rail shear (Iosipescu), and d) interlaminar shear tests.

are provided, as needed, in Section 4.3 along with the presentation of each specific FE model and the discussion of the results.

The interlaminar shear tests were performed in a *Form + Test Seidner* press, with a load capacity of 10 kN. The load was measured with the press built-in load cell and the displacements were measured with a *TML* displacement transducer, with a 25 mm stroke and 0.01 mm precision. The remaining tests were performed with an *Instron* universal test machine, model 1343, with a 250 kN load capacity and 100 mm stroke. The load and machine cross-head relative displacement were measured with the built-in transducers, while strains were measured with a video-extensometer.

In general, the results obtained in the material characterization tests presented relatively low coefficients of variation (CoV). More importantly, as expected, the tests performed in the different plates

showed similar results, both for strength and elasticity moduli, well within the expected experimental scatter, confirming that the plates have the same fibre architecture. Moreover, for I150-W specimens, the only tested in both tension and compression for both directions, the compressive and tensile elastic moduli in each direction presented similar values, showing that it is not necessary to consider different elastic moduli for compression and tension in the numerical modelling.

Table 1 presents three additional parameters to help with the calibration procedures and comparison with numerical results, namely the stiffness of the interlaminar shear tests (K_{13} and K_{23}) and the transverse tensile “yield” stress corresponding to the end of the proportionality limit ($S_{22,y}^+$). This last parameter was included due to the markedly non-linear behaviour observed from the early stages of these tests. The

end of the proportionality limit was defined when the experimental load consistently deviated from a linear response with the initial elastic modulus (a 0.1 MPa deviation was considered for this purpose).

4.2. Numerical modelling

4.2.1. Description of FE models

The FE models were developed with ABAQUS FE software package [23] and the proposed failure initiation and damage propagation models were implemented in a user material (UMAT) subroutine written in Fortran 77 [42]. Since the main objective of the present work is to develop a design methodology that can be applied to real engineering structural problems, the reduction of the computational costs of the analysis is a key aspect. In this context, the models used the simpler available solid elements, *i.e.* 8-node linear brick elements with reduced integration (C3D8R). Additionally, when possible, the number of elements was reduced taking advantage of symmetry conditions. With this regard, it should be mentioned that in the following sections, the figures of the models and meshes are portrayed as modelled, *i.e.* showing only the reduced models in accordance to the symmetry boundary conditions. On the other hand, when failure patterns are portrayed, they show the complete specimens, replicating the results of the reduced models over the symmetry planes, aiming at facilitating the understanding of the failure patterns.

Implicit analyses were conducted allowing for a maximum residual flux norm of 1%, except for severe discontinuity iterations, for which that value was increased to 5%. Boundary conditions were always defined as “hard” boundary conditions, by pinning the corresponding nodes.

Moreover, whenever necessary, contact formulations were prescribed, using the “Finite Sliding” formulation, with “Hard contact” properties for the normal behaviour and isotropic penalty properties for the tangential behaviour, with a friction coefficient of 0.1.

Specific details of each model, such as dimensions and boundary conditions, are presented and discussed in the following section.

Table 2 presents the mechanical properties used as input data list for each of the tested GFRP plates; the great majority were directly retrieved from coupon tests, namely the elastic and strength proper-

ties. As mentioned (*cf.* Section 4.1), similar elastic moduli were obtained in tension and compression (*cf.* Table 1), therefore the authors adopted the tensile elastic moduli as input. The properties that could not be directly retrieved from testing, duly identified in Table 2, were taken from the literature. In particular, the elastic modulus and tensile strength in the through-thickness direction (S_{33}^+ and E_{33} , respectively) were based on test results of Vallée [43], who tested 5 mm and 10 mm thick pultruded GFRP plates similar to those used in the present work (E-glass fibres, polyester matrix). Regarding the through-thickness compressive strength (S_{33}^-), to the authors’ best knowledge, no experimental data is available for pultruded GFRP materials. The few data available about the through-thickness compressive strength of 2D reinforced FRP laminates (with glass, carbon and aramid reinforcement), with thicknesses varying from 5 mm to 58 mm [44–46], ranges from ~140 MPa to ~900 MPa. Since in the present experiments through-thickness compressive failure was not observed, a conservative value of 90 MPa was adopted in the modelling. In this regard, it should be mentioned that: (i) the through-thickness compressive failure was not observed in the FE models, and thus the conservative value adopted did not govern the damage response; and (ii) the adoption of a larger value could lead to a significant increase of the interlaminar shear strength under moderate through-thickness compression, following a similar behaviour to that of the transverse plane (1–2), depicted in Fig. 2, which is not yet backed by experimental data on these particular materials.

On the other hand, it was not possible to determine the shear moduli and Poisson’s coefficients for planes 13 and 23, thereafter, for modelling purposes, the values obtained for plane 12 were adopted.

Regarding the tensile strength in the transverse direction, owing to the high non-linearity observed since the earlier stages of the tests (*cf.* Section 4.3.3), the authors adopted the “yield” stress (S_{22}^+) as input. Nonetheless, it should be mentioned that, with different calibration parameters, similar results to those presented in Section 4.3.3 could be produced.

Moreover, because it was not possible to perform all tests for the I150-F plate, the lacking input data, namely elastic moduli and strengths, were taken as those obtained for the I150-W plate (similar fibre architecture).

Table 3 presents the damage progression model parameters obtained from the calibration with the experimental tests (*cf.* Section 4.3). As expected, owing to the similar characteristics of both plates (fibre architecture and matrix), the calibration parameters obtained are the same for both plates, with the exception of $\varepsilon_{11,u}^+$, which was calibrated based on the longitudinal compressive strength obtained for the I150-W plate, but using the longitudinal tensile elastic modulus of the I150-F, and $\gamma_{23,u}^+$ as further discussed in Section 4.3.7. Additionally, for the transverse direction, since only the I150-W plate was tested, the corresponding parameters were assumed to be the same for the I150-F, as duly identified in Table 3. Moreover, since no through-thickness tests were performed and information in the literature is scarce, as mentioned *a priori*. During the tests presented herein, compressive through-thickness failure was never observed. Thereafter, in this case, conservatively, it was considered that no damage would occur in the progression stage, by setting $d_{3,max}^-$ to zero (0.001 was used as input value, *cf.* Table 3). Regarding through-thickness tensile failure, although it was not the primary failure mode in any of the tests performed, it might have had some influence in the post-failure behaviour, particularly for the transverse compression and interlaminar shear tests. On the other hand, since a limit strain could not be estimated, the constant stress state of the damage progression model (*cf.* Fig. 3) could not be used. Therefore, in this case, the damage progression stage (*cf.* Fig. 3) was used to attribute a severe and abrupt drop in the tensile through-thickness elastic modulus, namely by setting $d_{3,max}^+$ to 0.9 and m_3^+ to 100 (*cf.* Table 3).

Table 2
Input data for the FE models: mechanical properties of each GFRP plate.

Parameter	Unit	Input value	
		I150-W	I150-F
E_{11}	GPa	43.36	39.59
E_{22}		9.39	9.39 [†]
E_{33}		3.50*	3.50*
G_{12}		3.01	3.27
G_{13}		3.01**	3.27**
G_{23}		3.01**	3.27**
ν_{12}	–	0.23	0.29
ν_{13}		0.23**	0.29**
ν_{23}		0.30*	0.30*
S_{11}^+	MPa	387.97	353.47
S_{11}^-		461.94	461.94 [†]
S_{22}^{+1}		15.18	15.18 [†]
S_{22}^-		64.18	64.18 [†]
S_{33}^+		8.01*	8.01*
S_{33}^-		90.00*	90.00*
S_{12}		46.76	47.89
S_{13}		27.01	31.22
S_{23}		8.09	6.34

* Values estimated based on literature.

** Values obtained for plane 12 were adopted.

† Values obtained for I150-W.

Table 3

Input data for the FE models: damage progression parameters of each GFRP plate.

Parameter	Unit	Input value	
		I150-W	I150-F
$d_{1,max}^+$	–	0.001	0.001
$d_{1,max}^-$		0.001	0.001 [†]
$d_{2,max}^+$		0.55	0.55 [†]
$d_{2,max}^-$		0.5	0.5 [†]
$d_{3,max}^+$		0.9*	0.9*
$d_{3,max}^-$		0.001*	0.001*
$d_{4,max}$		0.95	0.95
$d_{5,max}$		0.5	0.5
$d_{6,max}$		0.5	0.5
m_1^+		10	10
m_1^-		10	10 [†]
m_2^+		1	1 [†]
m_2^-		10	10 [†]
m_3^+		100*	100*
m_3^-		100*	100*
m_4		5	5
m_5		1	1
m_6		100	100
$\varepsilon_{11,u}^+$		9.0×10^{-3}	9.0×10^{-3}
$\varepsilon_{11,u}^-$		1.1×10^{-2}	1.2×10^{-2}
$\varepsilon_{22,u}^+$		1.0×10^{-2}	$1.0 \times 10^{-2\ddagger}$
$\varepsilon_{22,u}^-$		1.0×10^{-2}	$1.0 \times 10^{-2\ddagger}$
$\varepsilon_{33,u}^+$		1*	1*
$\varepsilon_{33,u}^-$		1*	1*
$\gamma_{12,u}$		1	1
$\gamma_{13,u}$		8.5×10^{-3}	8.5×10^{-3}
$\gamma_{23,u}$		6.5×10^{-3}	4.1×10^{-3}
r_1^+		0.01	0.01
r_1^-		0.05	0.05 [†]
r_2^+		0.05	0.05 [†]
r_2^-		0.20	0.20
r_3^+		1*	1*
r_3^-		1*	1*
r_4		1	1
r_5		0.40	0.40
r_6		0.40	0.40
$\eta_i^{\pm}$		1×10^{-3}	1×10^{-3}

[†] Values obtained for I150-W.

* Values defined *a priori* due to the lack of experimental data.

4.2.2. Mesh dependency and sensitivity analysis

As shown by Bazant [19], using damage models based on fracture energy through stress–strain constitutive laws on FE analysis may lead to mesh dependency, in which the dissipated energy, after the peak load is achieved, tends to 0 as the mesh size decreases. In order to overcome this problem, that author proposed an artificial regularization of the fracture energy input based on the mesh size, which was later adopted in ABAQUS [23,24]. In the present model, however, since the decay of the elastic moduli is not a function of a fracture energy input, this problem does not occur. This can be readily shown by performing a similar test as that proposed by Lapczyk and Hurtado [24]. Thereafter, a specimen with dimensions of $120 \times 10 \times 10 \text{ mm}^3$, modelled with double-symmetry boundary conditions, was subjected to uniform tension in the transverse direction, by means of a uniform displacement applied to one of its extremity faces. The specimen was modelled with C3D8R cube elements with sizes of (i) 10 mm; (ii) 5 mm and; (iii) 2.5 mm, resulting in 12, 24 and 48 horizontal “slices”, as depicted in Fig. 7.

The input values of the I150-W material (cf. Tables 2 and 3) were considered, with the exception of one of the “slices”, for which the limit strains and distortions (which govern the initiation of the con-

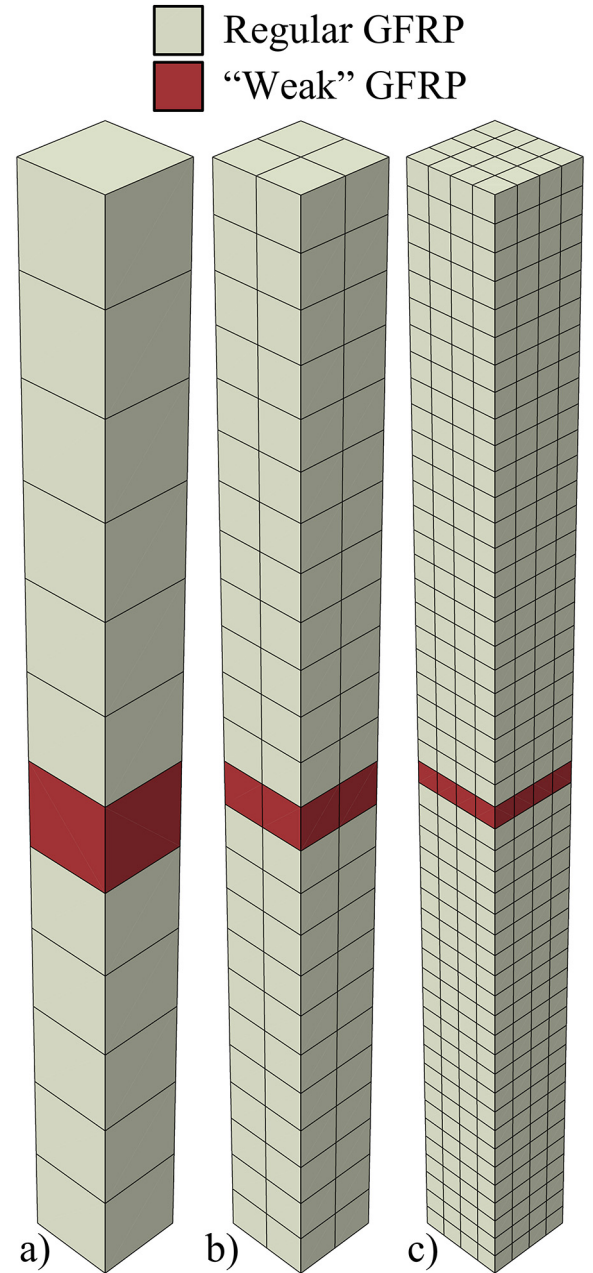


Fig. 7. Mesh dependency test – model with different mesh sizes: a) 10 mm (12 slices); b) 5 mm (24 slices) and; c) 2.5 mm (48 slices).

stant strength stage, cf. Fig. 3) were reduced by 10% in order to create a “weak” region, which would concentrate the damage. Moreover, this analysis was conducted with and without the regularization stage (cf. Fig. 5), by considering the regularization parameter (α) as 15 mm and 0 mm, respectively.

Fig. 8 compares the axial load vs. displacement curves obtained for the different mesh sizes, with and without regularization stage. When the regularization stage is not implemented (i.e. $\alpha = 0 \text{ mm}$), the same exact maximum load is attained with the different meshes, showing that the proposed damage progression model is not mesh dependent, unlike those based on fracture energies [19,24]. Thereafter, for uniform stress states, no stress regularization is required. On the other hand, with finer meshes, convergence problems occur after the peak load is attained. Conversely, when the regularization stage is taken

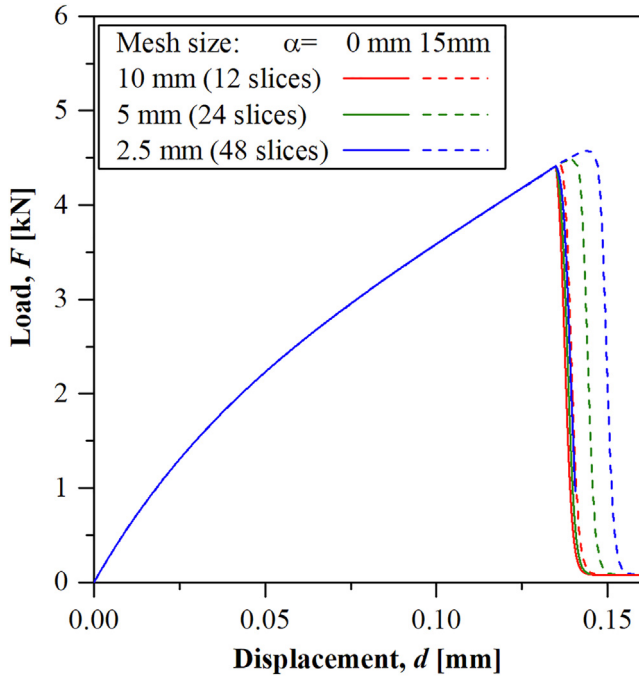


Fig. 8. Mesh dependency test – Load vs. displacement curves as a function of mesh size and regularization.

into account ($\alpha = 15$ mm), for the smaller mesh size (4 times smaller than the reference one), failure is slightly delayed, with a very slight increase in maximum load (+3.3%), while convergence in the post-peak descending branch is improved.

These results show that the proposed progression model does not have mesh dependency. When using the regularization stage (cf. Fig. 5), a small mesh sensitivity is found in this uniform stress state case. However, the introduction of the stress regularization stage is meant to avoid high mesh sensitivity in FE models where singularities or contacts exist. Moreover, albeit the precision of the model may be slightly affected by introducing the mesh regularization stage in these simple uniform stress state cases, most engineering problems require the use of contacts or singularities; therefore, the stress regularization stage will be used henceforth. Additionally, the influence of the regularization parameter (α) is further discussed in Section 4.3.1.1, while the need of adding the regularization stage is demonstrated in Part II [30].

4.3. Specific models, results and discussion

This section presents the modelling of each standardized coupon test, the results and discussion, regarding the comparison to experimental data, and the calibration of the damage progression parameters. The models presented herein can be divided in two main groups, namely those with (i) uniform and (ii) non-uniform stress states. The former group comprises the tensile and compressive tests in the longitudinal and transverse directions, while the latter includes the in-plane and interlaminar shear tests. For the cases with uniform stress states, the models included only the gauge length (triple symmetry boundary conditions were adopted in order to reduce computational costs) and the displacements were imposed uniformly to the extremities of the specimens, effectively imposing a uniform stress state, precisely what is intended with the standard test setups. Failure onset was promoted by adding a fictional imperfection, by means of a “weak” region, with an area of 1 mm^2 , which was extended through the thickness of the specimens, as exemplified in Fig. 9 for the longitudinal tensile test. The material adopted for the “weaker” regions was

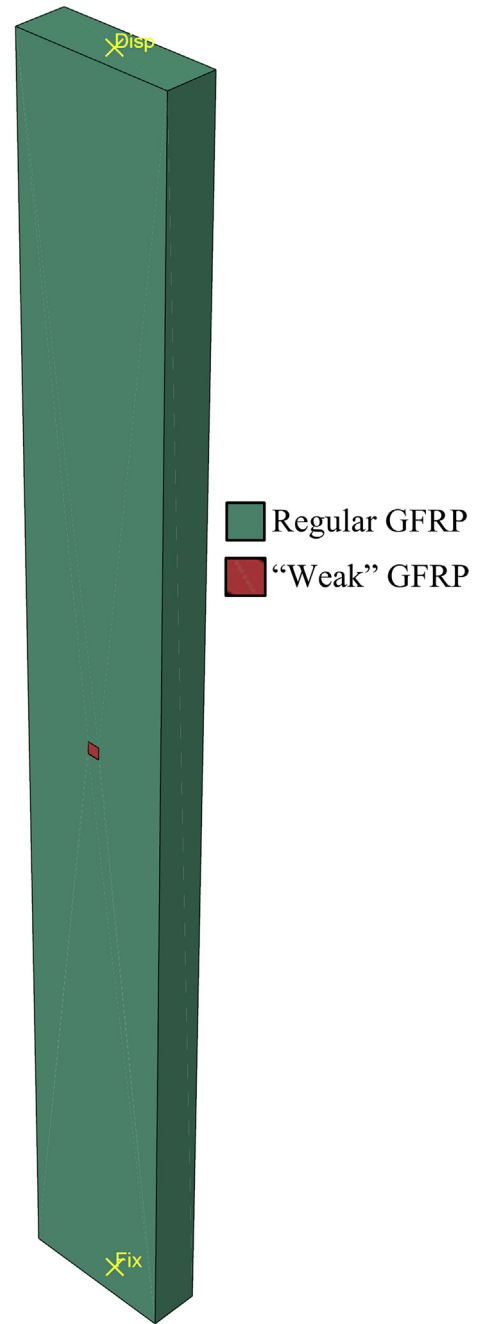


Fig. 9. Uniform stress state models with imperfection: longitudinal tensile test.

modelled with the properties given in Tables 2 and 3, with the exception of the limit strains ($\epsilon_{u,u}^{\pm}$ and $\gamma_{jk,u}$), which were reduced by 10%. Additionally, it should be mentioned that in the uniform stress state experiments, it was not always possible to measure the strains up-to and after failure, since in the brink of failure and subsequently the strain measurements become unreliable (the video-extensometer is no longer able to track the measurement points); thereafter, in the next sections, the numerical results are compared with both experimental stress vs. strain curves and load vs. cross-head displacement curves, as the post-failure behaviour was important for the calibration. Since the cross-head displacement measured by the test machine accounts for the whole test setup displacement, and not only for that of the test specimens, the relative displacement presented in those cases was fit-

ted to the numerical results to allow for a better comparison. Nevertheless, it should be stressed that the numerical and experimental values of stiffness are already compared by the measurement of the elastic moduli, based on the stress vs. strain results.

4.3.1. Longitudinal tensile tests

As mentioned (cf. Table 1), the longitudinal tensile tests were conducted according to EN ISO 527[37]. Fig. 10 presents the mesh details around the fictional imperfection of the model with three different global mesh sizes, (i) 1 mm, (ii) 0.5 mm and (iii) 0.25 mm. As depicted in Fig. 10, owing to the dimensions of the specimen and of the imperfec-

tion, it was not possible to create a regular mesh but rather a cuboid mesh.

The specimens tested in longitudinal tension presented linear behaviour up to brittle failure. Thereafter, in this case, the damage progression stage (cf. Fig. 3) can be readily ignored by setting $d_{1,max}^+$ to zero (a value of 0.001 was used as input, cf. Table 3). Moreover, the limit strain ε_{11}^+ can be estimated by the ratio between the longitudinal tensile strength (S_{11}^+) and the input longitudinal elasticity modulus (E_{11}).

The results obtained with the FE models are summarized in Table 4, showing an excellent agreement with the experimental results, both in terms of longitudinal strength and elastic modulus. Additionally,

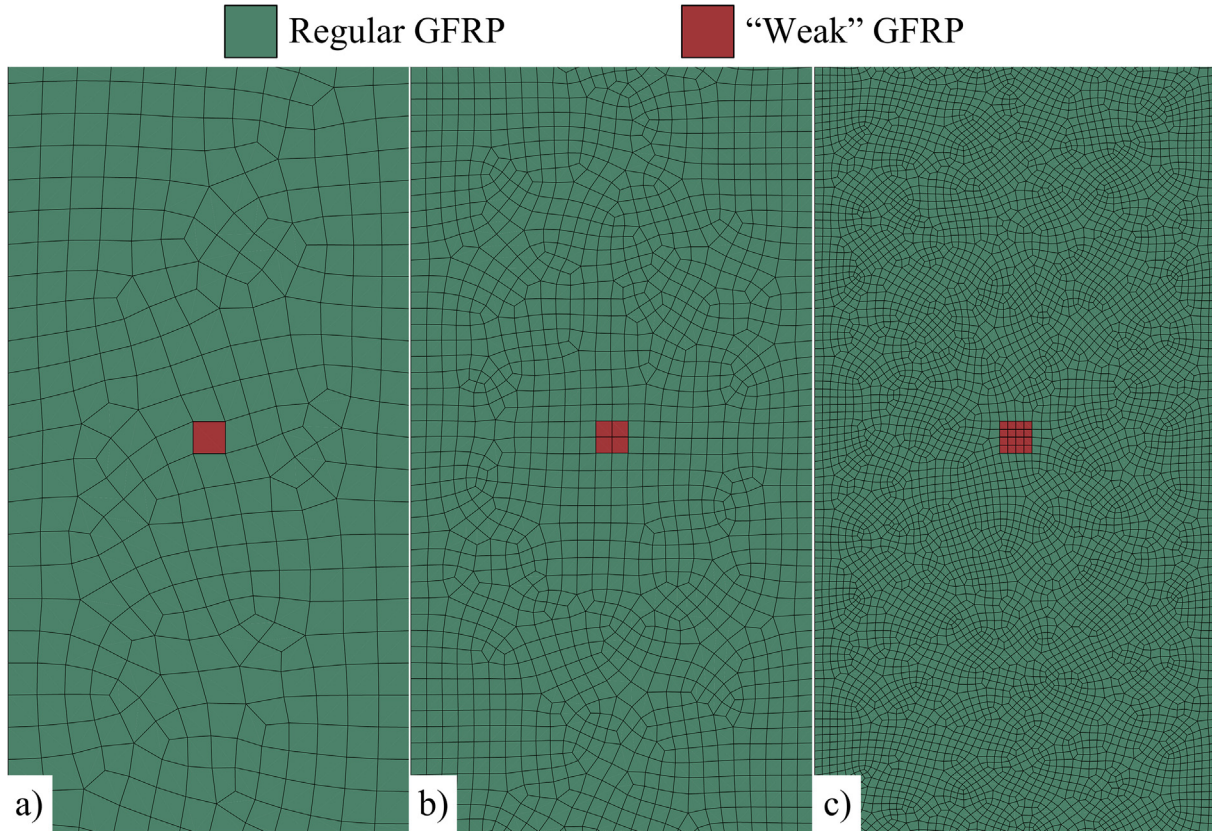


Fig. 10. Mesh details around the imperfection for the longitudinal tensile test model with global mesh sizes of: a) 1.00 mm; b) 0.50 mm; and c) 0.25 mm.

Table 4

Range of main results of the calibrated models, for different mesh sizes, and relative difference to experimental data (Δ).

Test	Property	Unit	I150-W		I150-F	
			Value	Δ	Value	Δ
Longitudinal tension	S_{11}^+	MPa	388.7 to 390.9	0.2% to 0.8%	354.9 to 356.8	0.4% to 0.9%
	E_{11}	GPa	43.4	0.0%	39.6	0.0%
Longitudinal compression	S_{11}^-	MPa	468.9 to 473.4	1.5% to 2.5%	–	–
	E_{11}	GPa	43.4	–3.4% ¹	–	–
Transverse tension	S_{22}^+	MPa	47.8 to 47.9	6.9% to 7.3%	–	–
	E_{22}	GPa	9.40	0.1%	–	–
Transverse compression	S_{22}^-	MPa	69.5 to 69.5	8.3%	–	–
	E_{22}	GPa	9.40	20.3% ¹	–	–
In-plane shear	S_{12}	MPa	42.9 to 45.8	–8.3 to –2.1%	44.9 to 45.5	–6.1% to –5.0%
	G_{12}	GPa	2.26 to 2.29	–24.5% to –23.8%	2.67 to 2.70	–18.3% to –17.6
Longitudinal inter-laminar shear	S_{13}	MPa	23.6 to 26.6	–12.7% to –1.6%	28.2 to 31.5	–9.7% to 0.9%
	K_{13}	MPa/mm	63.9 to 64.9	–1.9% to –0.4%	67.7 to 68.9	–1.3% to 0.4%
Transverse inter-laminar shear	S_{23}	MPa	7.47 to 8.68	–7.7% to 7.2%	6.74 to 6.83	–0.5% to –1.9%
	K_{23}	MPa/mm	24.7 to 25.7	–7.4% to –4.0%	23.6 to 24.3	–6.7% to –3.7%

¹ Comparing to the correspondent compressive experimental modulus.

Fig. 11a and b compare the numerical and experimental results in terms of load vs. displacement, while Fig. 11c present a similar comparison in terms of longitudinal stress vs. strain, for the I150-W and I150-F plates, respectively. The comparison of numerical and experimental stress vs. strain results allows assessing the quality of the simulation in terms of stiffness. However, as mentioned earlier, since the strain measurements are unreliable in the brink of failure and beyond, the load vs. displacement curves are useful to compare the experimental and numerical ultimate strength and post-failure behaviour, in particular, the residual loads.

The failure mode predicted by the model was identical to that observed in the experiments, with failure initiating in the imperfection and progressing to the lateral edges of the specimens, as exemplified in Fig. 12 for the model of the I150-W plate with global mesh size of 1 mm. As mentioned (cf. Section 4.2), for a better understanding of the failure pattern, the numerical results are replicated over the symmetry planes and the whole specimen is portrayed.

Regarding the mesh size sensitivity, the results show that it has little impact in terms of the predicted strength, with maximum differences between meshes under 0.6% – all numerical results are in good agreement with test data (cf. Fig. 11a and b). Yet, it can be seen that finer meshes seem to slightly delay the load decay; this effect is related to the use of the regularization stage (cf. Fig. 5) – as mentioned,

this is a concession to avoid mesh sensitivity in models with singularities that lead to a small loss of accuracy in uniform stress state cases (cf. Section 4.2.1). Overall, these results show that the proposed damage progression model, duly calibrated, is well able to reproduce the behaviour observed in the experiments.

4.3.1.1. Effect of the regularization factor (α). In order to avoid mesh sensitivity issues when singularities exist, the transition between the damage progression and constant stress stages (cf. Fig. 5) is smoothed by means of a regularization stage, which depends on the mesh size and on a regularization factor (α), cf. Section 3. Thereafter, a sensitivity analysis of the regularization factor was performed, based on the longitudinal tensile test of the I150-W plate with a global mesh size of 1 mm.

Fig. 13 presents the experimental load vs. displacement curves and compares them with numerical results obtained with regularization factors (α) ranging from 0 to 150 mm. The results show that the regularization factor has little effect on the maximum load attained, except for very low values ($\alpha < 5$ mm). On the other hand, large values ($\alpha > 20$ mm) seem to excessively delay the load decay. Based on these results, the regularization factor was set to 15 mm in all analyses. Again, it should be stressed that this parameter does not have a physical meaning and it is used only to avoid mesh sensitivity issues on FE

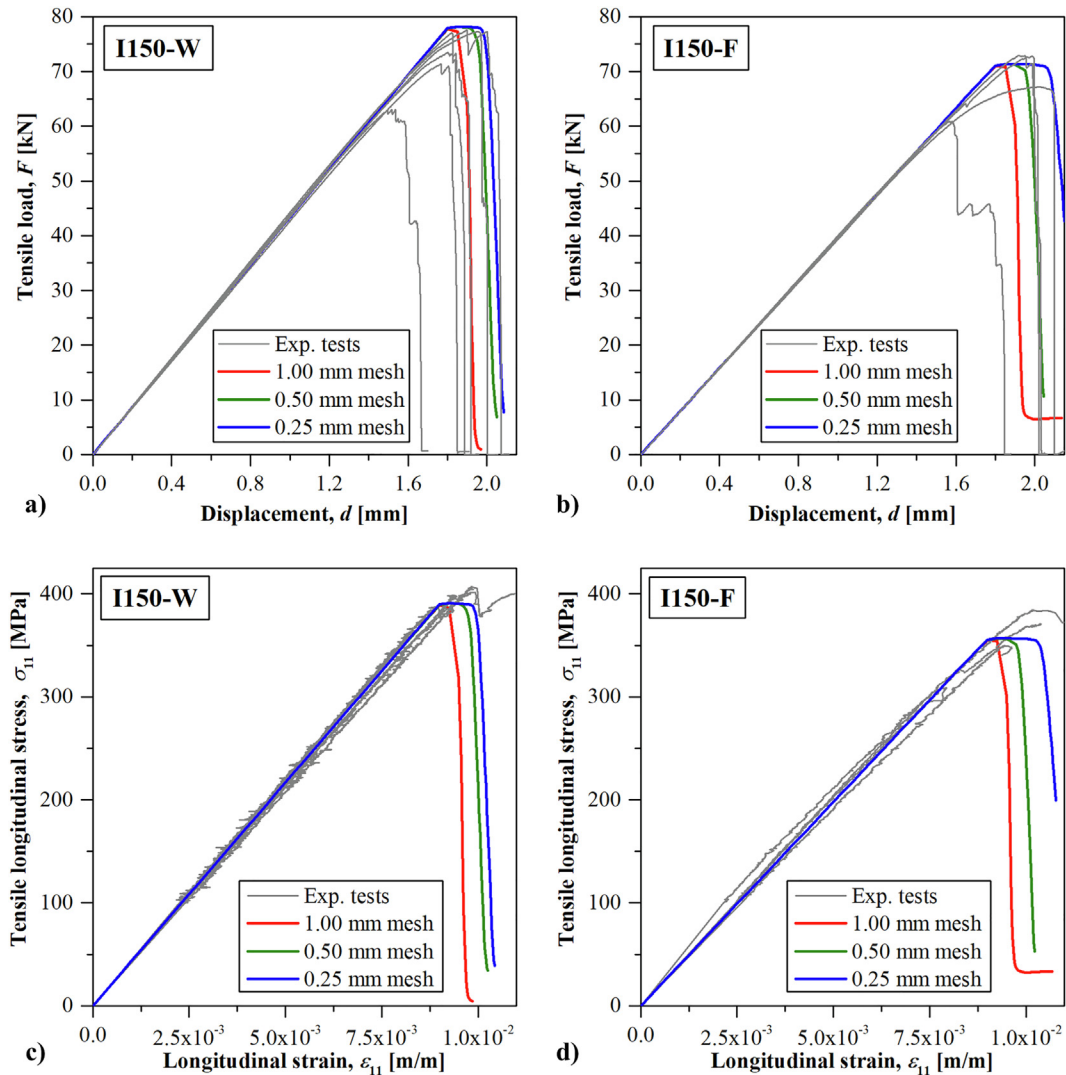


Fig. 11. Longitudinal tensile tests experimental vs. numerical results: load vs. displacement curves for a) I150-W and b) I150-F plates and; stress vs. strain curves for c) I150-W and d) I150-F plates.

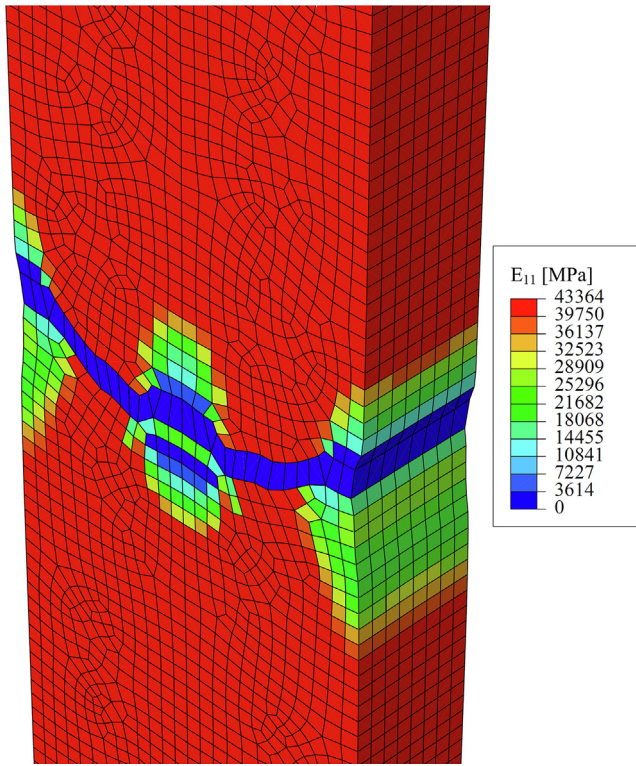


Fig. 12. Final damage (in terms of E_{11}) of the longitudinal tensile test model of I150-W plate with global mesh size of 1 mm.

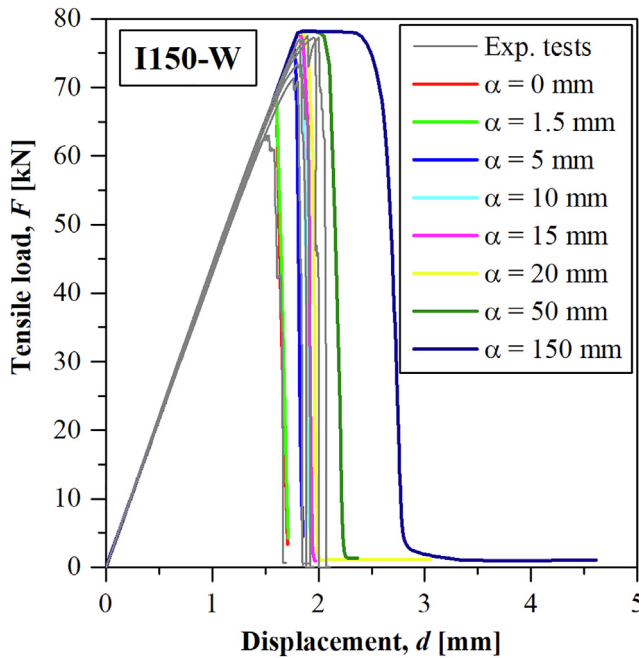


Fig. 13. Sensitivity analysis of the regularization parameter (α) for the longitudinal tensile test: load vs. displacement curves of I150-W plate.

analysis. Moreover, since it is a non-dimensional parameter, it is equally applied for all load directions.

4.3.2. Longitudinal compressive tests

The compressive tests were performed using a combined loading compression (CLC) test apparatus (cf. Fig. 6b), following ASTM-

D6641 [38] (cf. Table 1). The models developed followed the simplifications specified earlier, namely, by modelling only the gauge length and using triple symmetry boundary conditions. In this case, owing to the dimensions of the specimens, the 1 mm² imperfection was slightly deviated from the centre in order to produce a regular mesh, as depicted in Fig. 14 for global mesh sizes of 1 mm, 0.5 mm and 0.25 mm.

The stress vs. strain behaviour obtained in the experimental longitudinal compressive tests was similar to that obtained in the tensile tests, namely a linear behaviour up to brittle failure. Thereafter, $d_{1,max}^-$ was also set to zero and the limit strains ($\epsilon_{11,u}^-$) were estimated from the ratio between the longitudinal compressive strength (S_{11}^-) and the longitudinal elasticity modulus (E_{11}) used as input. Since this test was performed only for web specimens, the strength adopted for the flanges was that retrieved from the I150-W specimens.

Again, the numerical results (summarized in Table 4) show an excellent agreement with the test data. Fig. 15a and b show the comparison between numerical and experimental load vs. displacement and stress vs. strain curves, respectively, depicting the ability of the proposed model to simulate the experimental behaviour. Moreover, regarding the mesh sensitivity, similar conclusions to those of the longitudinal tests can be drawn, particularly a very small influence on the maximum load (relative differences below 1%) and a larger influence on the post-failure behaviour, with finer meshes slightly delaying the descending branch after the peak load.

4.3.3. Transverse tensile tests

The transverse tensile tests were performed following the recommendations of EN ISO 527 [37]. However, as discussed (cf. Section 4.1), due to the geometry of the I-shaped profiles, it was only possible to conduct tests for web specimens (I150-W) and with a shorter length than recommended by the standard, namely, with 122 mm and 82 mm of total and gauge lengths, respectively. The mesh details around the imperfection are very similar to those of the longitudinal tensile modes (cf. Fig. 10).

Fig. 16a and b compare the experimental and numerical load vs. displacement and stress vs. strain curves, respectively. Unlike what was observed for the longitudinal tensile and compressive tests, the transverse tensile test specimens presented a bi-linear behaviour. In fact, this type of (expected) bi-linear tensile behaviour, very similar to that reported for polyester filaments (e.g. [47]), is governed by the resin matrix of the GFRP. Thereafter, in this case, the damage progression stage (cf. Fig. 3) has an important role in the accurate modelling of the material behaviour up to (brittle) failure. The results show that the proposed model, duly calibrated, has the ability to simulate the progressive stiffness loss, as depicted in Fig. 16b, as well as the ultimate strength (cf. Table 4), which, once again, was not influenced by the mesh size – the maximum relative difference of maximum loads for different mesh sizes was below 0.4%.

4.3.4. Transverse compressive tests

The transverse compressive tests were performed following ASTM-D6641 [38]. However, owing to the aforementioned dimensional limitations (cf. Section 4.1), they were only performed for web specimens (I150-W) and with a shorter length than that indicated in the standard, namely 123 mm and 16 mm of total and gauge lengths, respectively. Thereafter, the specimens were loaded only by the grips and not by the end plates. Notwithstanding, the initial failure mode of the specimens was compressive, along the free length, and, therefore, the results were considered valid.

As depicted in Fig. 17a, the load vs. displacement curves reflected a linear behaviour up to brink of failure, when a non-linear reduction of stiffness occurred prior to failure, which then involved a sudden load reduction. After failure, the specimens were able to maintain a considerable residual strength, between ~40% and ~70%. This somewhat

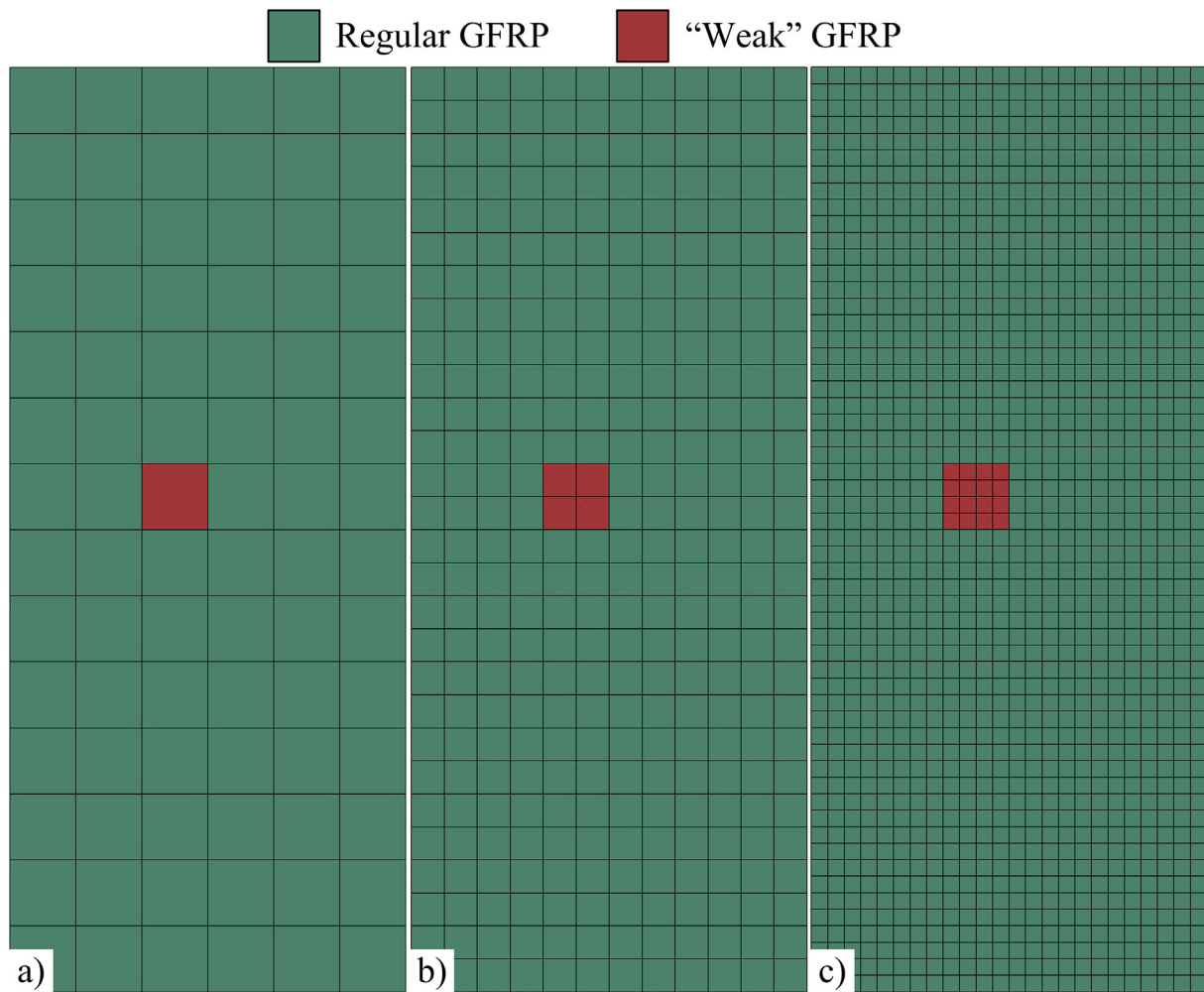


Fig. 14. Longitudinal compressive tensile test – mesh details for the model with global mesh sizes of: a) 1.00 mm; b) 0.50 mm; and c) 0.25 mm.

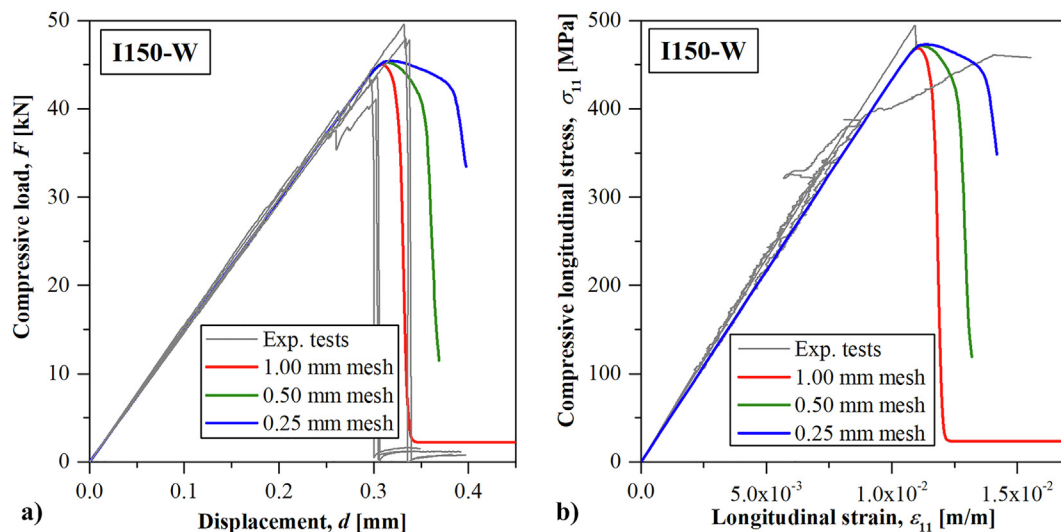


Fig. 15. Longitudinal compressive test – experimental vs. numerical results for the I-150-W plate: a) load vs. displacement and; b) stress vs. strain curves.

unexpected high residual strength raised concerns regarding the characterization of the post-failure behaviour in transverse compression of the laminates using the CLC test, which may be strongly influenced by this particular setup, in particular the reduced gauge length of the

specimens and the high constraints of the grips, which may have promoted crushing failure instead of “free” compressive failure.

In this context, additional compressive tests were conducted for the transverse direction following ASTM D 695 [41]. Although these tests

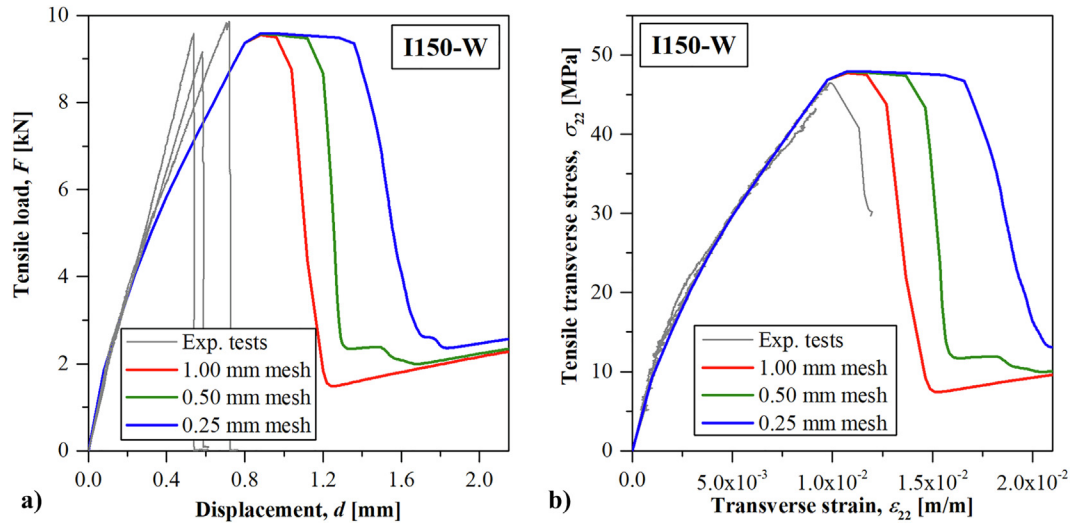


Fig. 16. Transverse tensile test – experimental vs. numerical results for the I-150-W plate: a) load vs. displacement and; b) stress vs. strain curves.

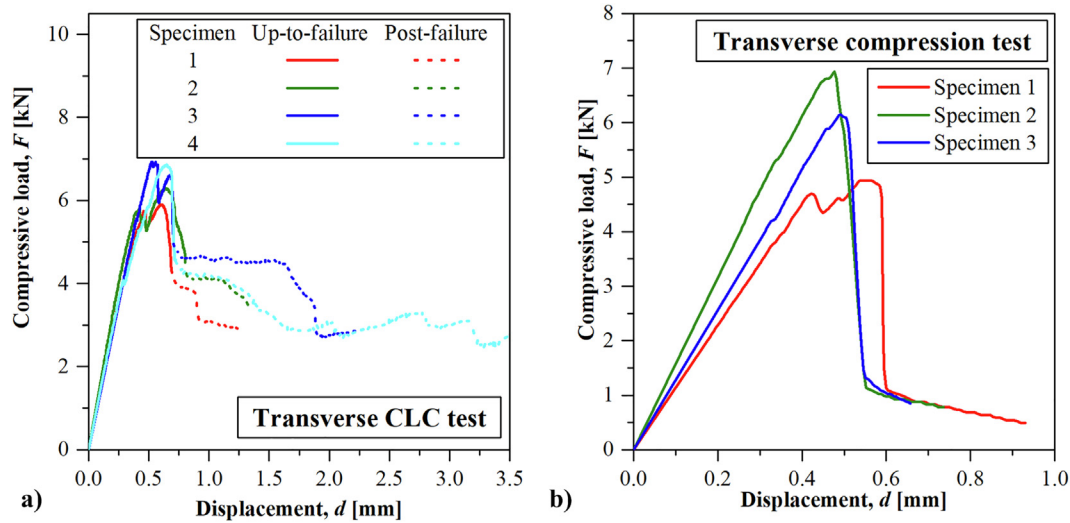


Fig. 17. Transverse compression test – experimental load vs. displacement curves for the I150-W plate following: a) ASTM-D6641 [38] (CLC) and b) ASTM D 695 [41].

are known to be less reliable than those of ASTM-D6641 [38], often underestimating the compressive strength [48], in this case they were used to clarify the aforementioned concerns regarding the post-failure behaviour. Fig. 17b presents the experimental load vs. displacement curves of the I150-W plate, presenting a residual strength ranging from 15% to 20%, which progressively decreased with the displacement increase – these figures confirm the overestimation of residual strength by the CLC tests. It should be noted that the average strength and elastic modulus obtained in these tests, 45.0 ± 7.3 MPa and 2.94 ± 0.53 GPa, respectively, were considerably lower than those obtained with the CLC tests, as expected, namely –30% and –64%, respectively.

Due to the aforementioned concerns, the authors decided to use the curves obtained with the CLC test to calibrate the behaviour up to failure, including the damage progression stage (cf. Fig. 3). After that stage, a 20% constant stress was adopted, based on the results of the ASTM D 695 [41] compression tests, resulting in the adoption of $r_2 = 0.2$ (cf. Table 3).

The FE model developed was very similar to that used for the longitudinal compressive tests, varying only in length and material orientation, and the mesh details are very similar to those presented in

Fig. 14. Fig. 18a and b show the experimental and numerical load vs. displacement and stress vs. strain curves, respectively. In this case, owing to the aforementioned limitations of the video extensometer readings in the brink of and after failure, the post-failure behaviour was assessed mainly through the load vs. displacement results – Fig. 18a shows that a good agreement was obtained up to failure. Unlike the previous models, in this case, the mesh size had absolutely no effect on the maximum load, with the different meshes presenting the same results. This is due to the fact that the maximum load is governed by the damage progression stage and not by the constant stress stage (cf. Fig. 5). Moreover, in the previous models, the descending branch of the load vs. displacement curves was delayed by the finer meshes, which were homothetic to each other. In this case, however, the descending load paths deviate considerably, with the finer meshes delaying the stage of constant stress. This difference is due to the fact that, in this case, owing to the input material properties, several failure modes occur concomitantly, in particular (i) transverse compression, (ii) in-plane shear, and (ii) through-thickness tensile failure, similarly to what was observed in the CLC tests. In fact, for the finer mesh, axial failure did not concentrate on a single element alignment stemming

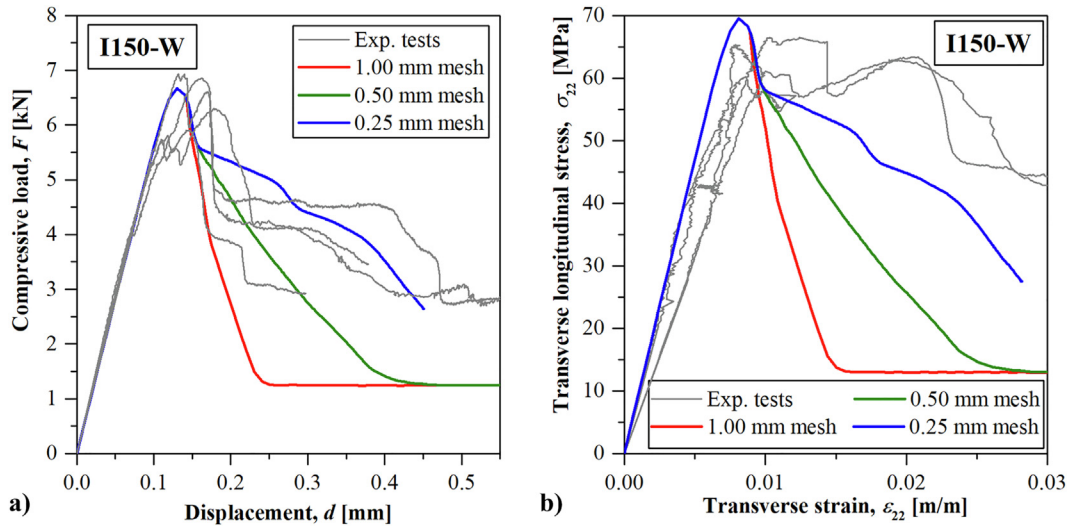


Fig. 18. Transverse compressive test – experimental vs. numerical results for the I-150-W plate: a) load vs. displacement and; b) stress vs. strain curves.

from the imperfection, as occurred in the previous cases (cf. Fig. 12), but spread along the height of the specimens as the in-plane and through-thickness tensile failure. In this regard, it should be stressed that, the numerical simulations aimed at evaluating uniform stress states, thus only the gauge length was modelled. In this case, the shorter gauge length used in transverse CLC experimental tests may not be sufficient to produce uniform stresses (according to Saint Venant's principle) and, therefore, the grips, not included in the models, may have significantly influenced the behaviour within the gauge length. Fig. 19 presents the damage evolution for the 0.25 mm mesh in terms of the transverse, through-thickness and in-plane distortion moduli (E_{22} , E_{33} and G_{12} , respectively), showing that up to the maximum load ($F = 6.7$ kN) the behaviour is governed by the damage progression stage, with the damage of the in-plane shear modulus evolving faster than that of the transverse elastic modulus (which governs the behaviour). For displacement above 0.15 mm, the constant stress stage is activated, as depicted in Fig. 19, with the transverse elastic modulus damage progressing faster than that of the in-plane shear modulus in the imperfection section. Nevertheless, for displacements higher than 0.20 mm, for which tensile damage along the through-thickness direction already develops, transverse compressive failure accumulates in two sections in the vicinity (but outside) of the imperfection, finally concentrating in one of them (the upper one) at the end of the simulation ($d = 0.45$ mm).

4.3.5. In-plane shear tests

The in-plane shear tests were conducted according to ASTM-D5379 [39], by testing V-notched specimens retrieved from both the web (I150-W) and flange (I150-F) plates. The FE simulations included three global mesh sizes, namely 2 mm, 1 mm and 0.5 mm, all depicted in Fig. 20. In order to reduce computational costs, a single longitudinal symmetry boundary condition was applied at mid-thickness of the specimens. In the test setup, specimens were restrained and loaded by a pair of grips at each side of the notch (cf. Fig. 6c). In the FE models, these grips are represented by the direct application of the displacement or support (pinned) at the “compression” grips (top right and bottom left, respectively, cf. Fig. 20), or by steel parts ($E_s = 200$ GPa and $\nu_s = 0.3$) at the “tension” grips (bottom right and top left, cf. Fig. 20) to avoid the development of unrealistic tensile stresses at these zones. These steel parts interact with the GFRP specimen by means of a compressive only “hard contact” surface interaction.

Fig. 21a and b compare experimental and numerical results for I150-W and I150-F specimens, respectively. The results show that

the FE models are well able to reproduce experimental results, particularly up to failure. In this respect, owing to the stroke limitations of the test device, it was not possible to test the specimens up to complete failure. Therefore, the residual strength stage (cf. Fig. 2) was not used in the calibration of the damage progression model; the authors adopted virtually unlimited ultimate strains ($\gamma_{12,u} = 1$, cf. Table 3). Additionally, the same damage progression parameters were adopted for both plates ($d_{A,max} = 0.95$ and $m_4 = 0.5$, cf. Table 4), with the difference of results stemming only from the elasticity and strength input values of each plate, namely G_{12} and S_{12} , cf. Table 2 (according to test results).

Regarding the mesh sensitivity of these models, as depicted in Fig. 21a and b, the mesh size has influence on both the maximum stress and the post-failure behaviour. For the simulation of the I150-W plate, the coarser mesh presented the less accurate strength estimates (-8% w.r.t. test data), while the intermediate and finer meshes provided similar predictions (-2% and -3% , respectively). All meshes presented post-failure paths within the experimental scatter, with the finer mesh presenting the most abrupt minute load reduction, after the peak strength. For the I150-F plate, similar strength estimates were obtained with all meshes, with relative differences to test results ranging from -5% to -6% . In the post-failure stage, the agreement between numerical and test results is worse, with the finer mesh presenting a steep load reduction after the peak strength, while the intermediate mesh was unable to converge after that point.

Regarding the failure pattern (another important issue when assessing the quality of the modelling), Fig. 22 compares experimental and numerical results for the I150-W plate with 1 mm mesh size, as an example. The FE models predicted the failure pattern correctly, namely capturing the compressive (crushing) failure caused by the grips at the onset of the notch and a well-defined vertical shear fracture along the notch, both clearly visible in Fig. 22a.

4.3.6. Longitudinal interlaminar shear tests

The interlaminar shear tests, both longitudinal and transverse directions, were performed following the recommendations of ASTM-D2344 [40], therefore the interlaminar stress estimates were obtained as recommended in that document. As expected, failure of all specimens from both web (I150-W) and flange (I150-F) plates occurred due to interlaminar shear. Similarly to the previous cases, the FE models of these tests were developed with symmetry boundary conditions in order to reduce computational costs, namely in the longitudinal and transverse directions. Moreover, the outer

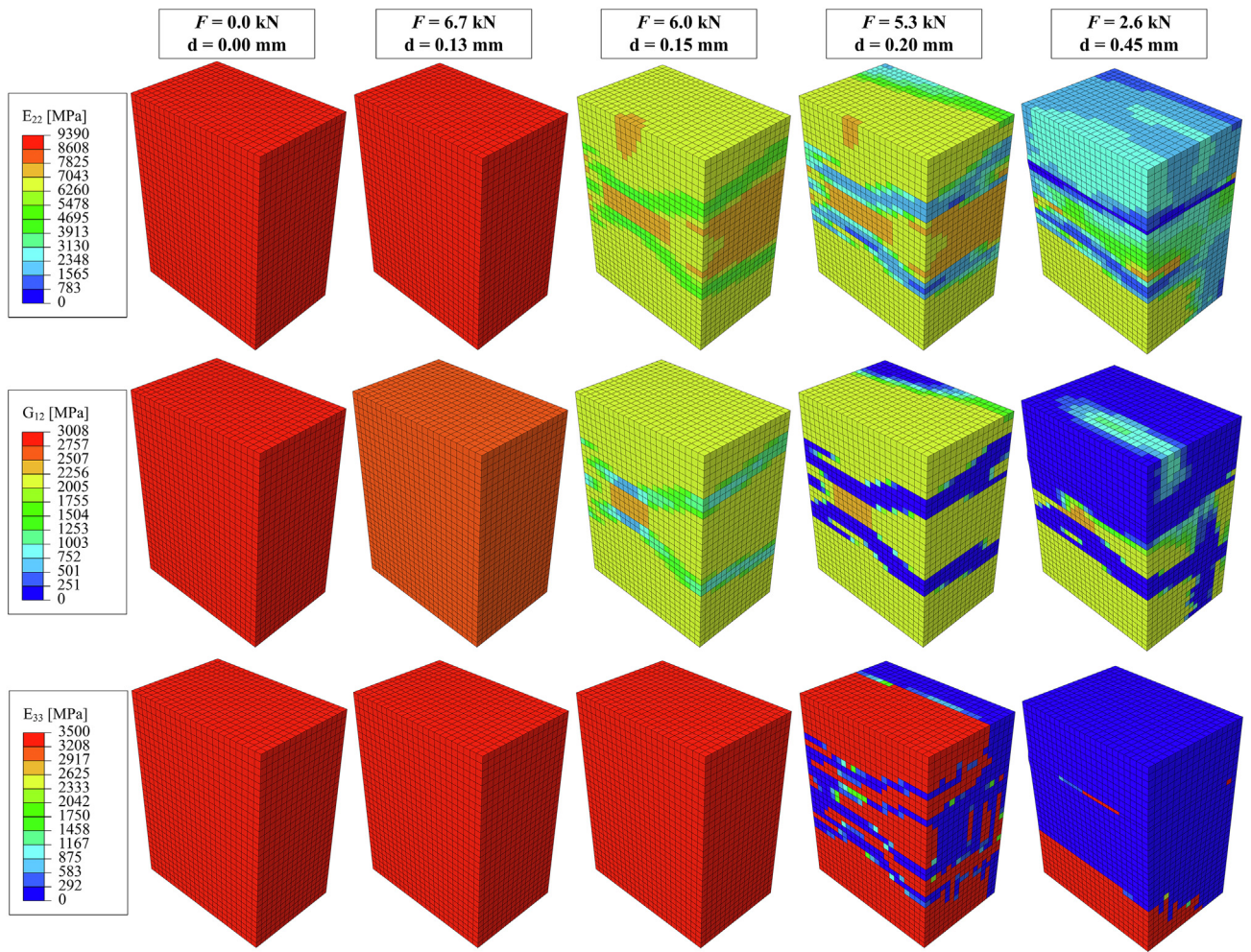


Fig. 19. Transverse compressive test – FE damage evolution, in terms of E_{22} , G_{12} and E_{33} , of the I150-W plate with global mesh size of 0.25 mm.

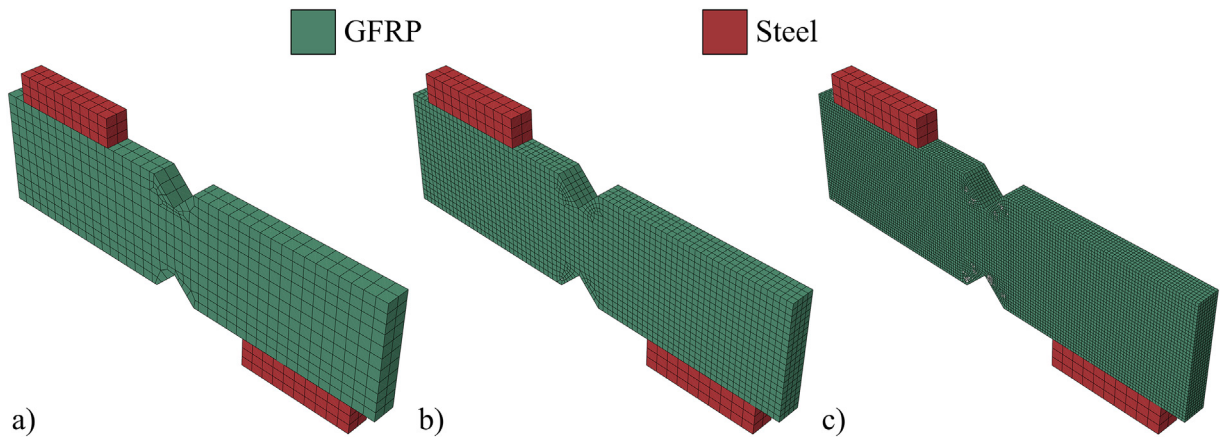


Fig. 20. In-plane shear test – model overview with global mesh sizes of: a) 2.0 mm; b) 1.0 mm; and c) 0.5 mm.

boundary conditions of the tests were simulated by steel parts ($E_s = 200$ GPa and $\nu_s = 0.3$) that interact with the FRP material through “hard contact” surface interactions, like in the in-plane shear test models (cf. Section 4.3.5). In this case, since the steel support/loading parts are cylindrical, they were finely meshed to reproduce their shape, with a constant horizontal distance of 0.125 mm between nodes; and the contact surfaces of the GFRP parts were

meshed to match those nodes in order to avoid unrealistic distortions of the GFRP material in the contact areas, as shown in Fig. 23 for global mesh sizes of 2 mm, 1 mm and 0.5 mm. This resulted in meshes with stretched parallelepipeds, instead of the (ideal) cubes, in the contact zones; however, this concession allowed for a significant reduction of the computational costs when compared to the use of a very fine mesh (0.125 mm sided cubes) in the contact zones. More-

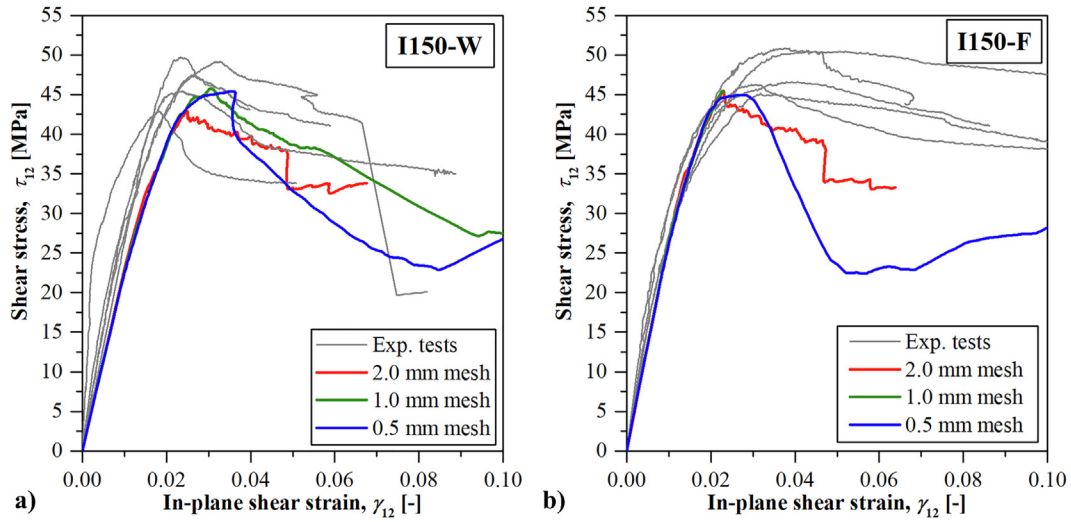


Fig. 21. In-plane shear test – experimental vs. numerical stress vs. strain curves for the: a) I150-W and; b) I150-F plates.

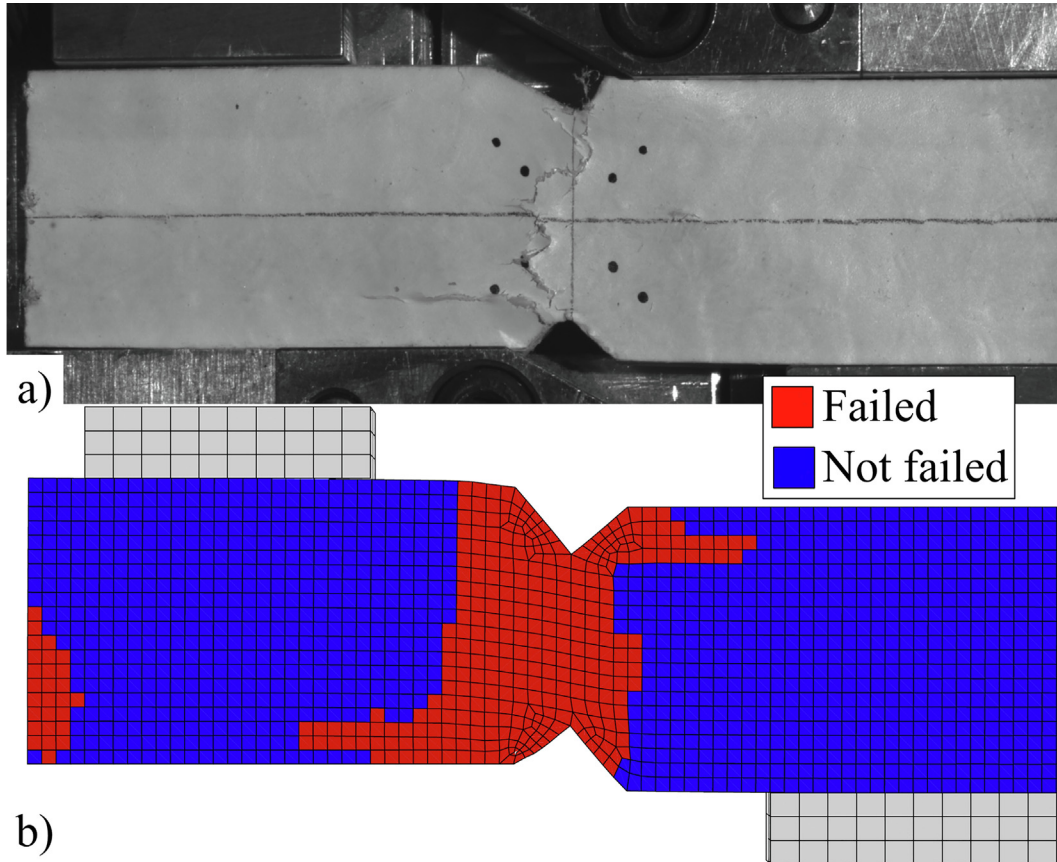


Fig. 22. In-plane shear test of I150-W plate – Final final failure pattern: a) experimental specimen and b) FE model (1 mm global mesh).

over, it should be mentioned that a few cases were tested with this type of regular cuboid mesh, providing very similar results to those obtained with the meshes presented in Fig. 23.

Fig. 24a and b present experimental and numerical interlaminar stress vs. vertical displacements for both plates. Once again, the results show the ability of the calibrated models (using the same relevant calibration parameters for both plates, cf. Table 3) to simulate the behaviour observed in the experiments. In particular, the onset of failure was correctly predicted with relative differences to (measured) mean

values of interlaminar strengths ranging from -13% to -10% for the coarser mesh and -4% to $+1\%$ for the finer mesh, cf. Table 4. Furthermore, the post-failure behaviour, characterized by the progression of interlaminar shear failure, was remarkably well captured up to ~ 2 mm of vertical displacement, particularly for the finer meshes; subsequently, severe shear damage (G_{13}) developed in one or more entire sections, near the supports, leading to very high “local” displacements – after this point, the simulation results are no longer representative of the experiments.

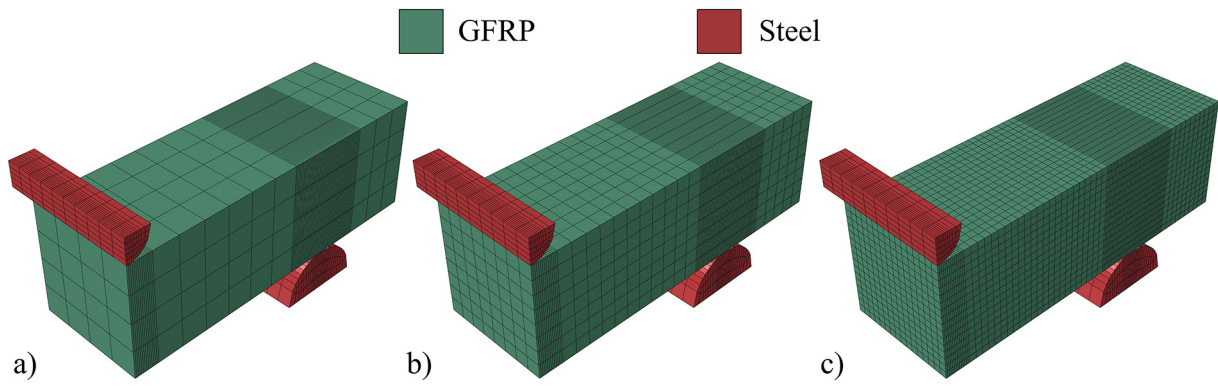


Fig. 23. Interlaminar shear test – model overview with global mesh sizes of: a) 2.0 mm; b) 1.0 mm; and c) 0.5 mm.

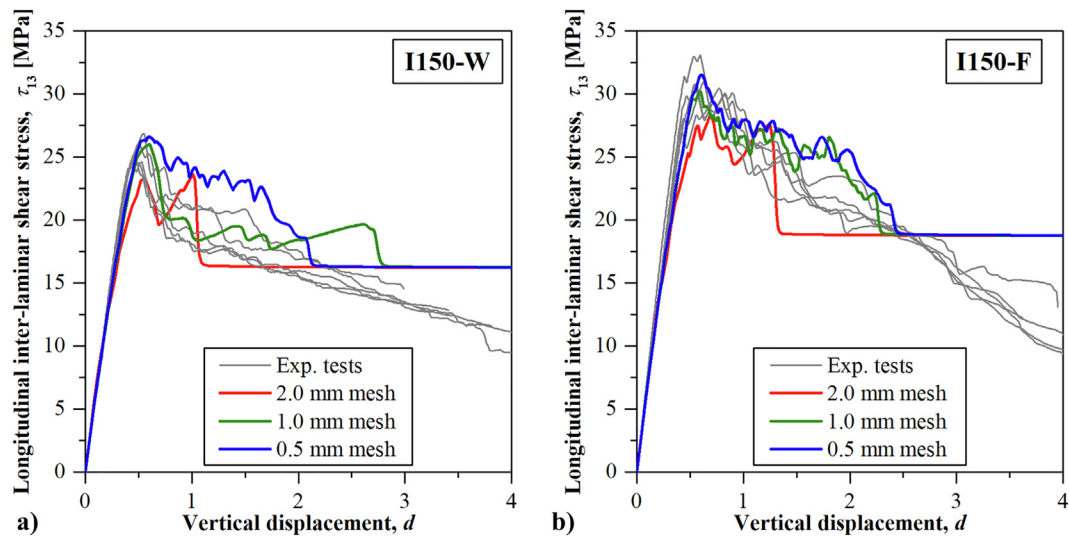


Fig. 24. Longitudinal interlaminar test – experimental vs. numerical stress vs. vertical displacement curves for the: a) I150-W, and b) I150-F plates.

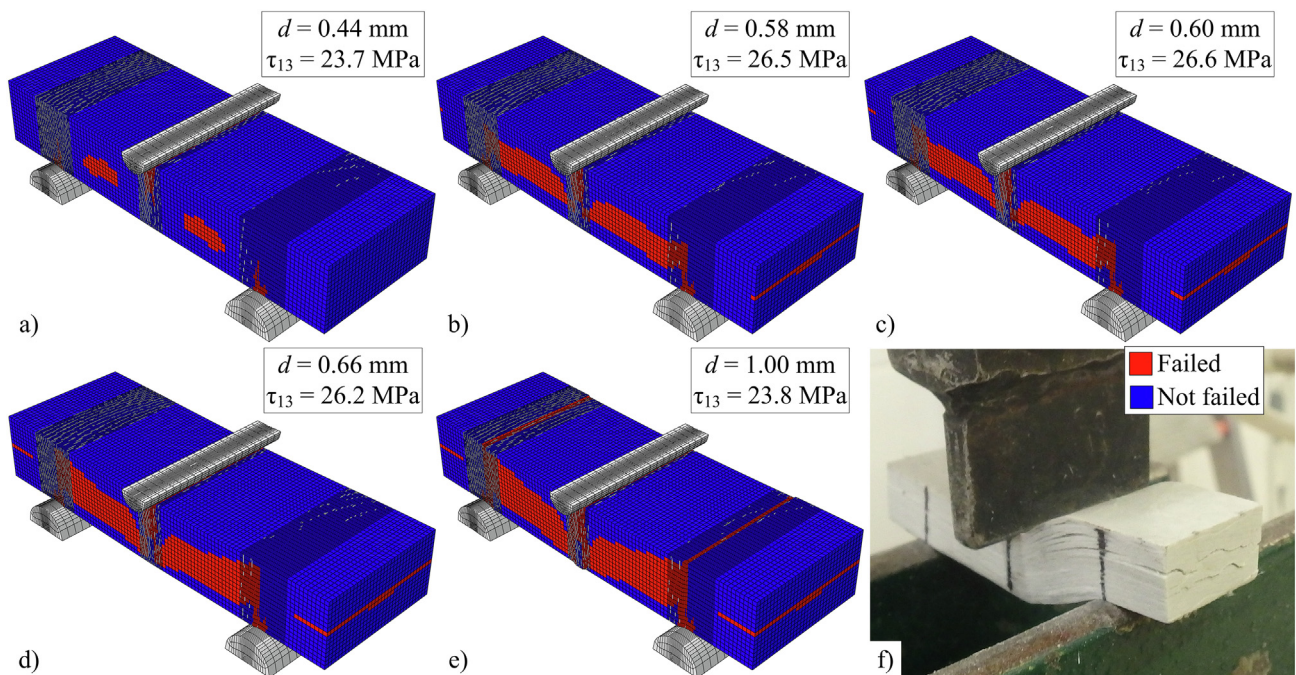


Fig. 25. Longitudinal interlaminar shear test – failure pattern progression predicted by the FE model (for plate I150-W, 0.5 mm global mesh), for vertical displacements of: a) 0.44 mm; b) 0.58 mm; c) 0.60 mm (maximum load); d) 0.66 mm; e) 1.00 mm; and f) final aspect of experimental specimen.

As depicted in Fig. 25, the numerical models also presented a remarkable agreement with the experiments in terms of failure pattern progression: failure was simulated to progress longitudinally until a damage surface between supports is reached (Fig. 25a–c), with additional occurrence of through-thickness tensile rupture at the free end of the specimen. As the vertical displacement is increased, further failure surfaces develop in-between the supports (Fig. 25d–e), similarly to the experiments (Fig. 25f).

4.3.7. Transverse interlaminar shear tests

The transverse interlaminar shear tests were also performed for specimens taken from the web (I150-W) and flange (I150-F) plates. Unlike what occurred for the specimens tested in the longitudinal direction (cf. Section 4.3.6), in the transverse interlaminar shear tests failure occurred due to in-plane shear (as opposed to interlaminar shear) in the 2–3 plane, in-between supports. Moreover, unlike the previous tests, failure was asymmetric, occurring in only one side of

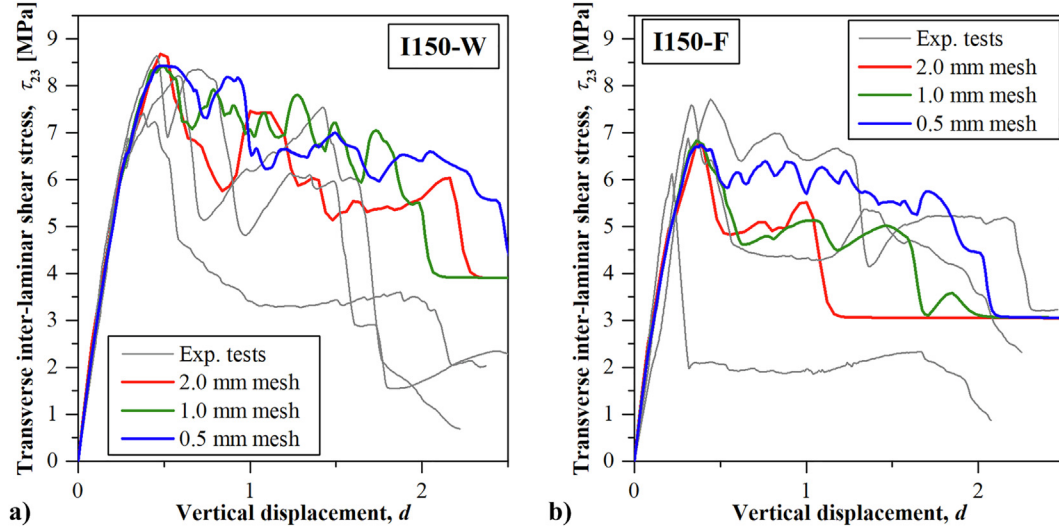


Fig. 26. Transverse interlaminar test – experimental vs. numerical longitudinal interlaminar shear stress vs. vertical displacement curves for the: a) I150-W and; b) I150-F plates.

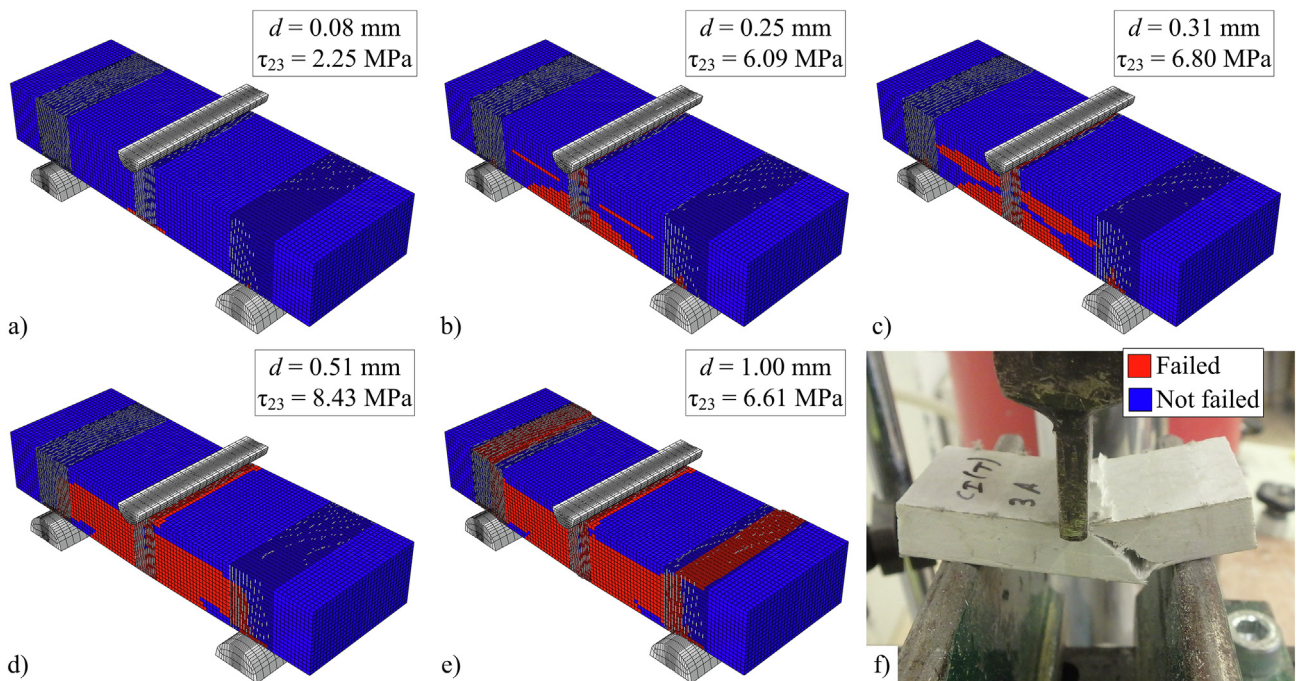


Fig. 27. Transverse interlaminar shear test – failure pattern progression predicted by the FE model (for plate I150-W, 0.5 mm global mesh), for vertical displacements of: a) 0.08 mm; b) 0.25 mm; c) 0.31 mm; d) 0.51 mm (maximum load); e) 1.00 mm; and f) final aspect of experimental specimen.

the specimens, presumably due to the presence of imperfections. Additionally, compressive crushing in the transverse direction under the load application point was also observed.

The FE model and meshes used to simulate these tests were identical to those used for the longitudinal direction (cf. Fig. 23), only the material orientation was changed. Fig. 26a and b compare experimental and numerical longitudinal interlaminar shear stress vs. displacement curves for the I150-W and I150-F plates, respectively. Once again, these results show that the damage progression model proposed is able to predict the initial behaviour and maximum stress with good accuracy for both plates, with relative differences to test data ranging from -8% to -1% in terms of strength (cf. Table 4). It should be mentioned that this was the only case in which the calibration of the damage model differed between the two plates: different transverse interlaminar limit shear strains ($\gamma_{23,u}$) were considered, owing to a significant difference in the experimental average strength (28% higher in plate I150-W, cf. Table 1); these differences were attributed to the fact that part of the flange specimens were taken from the web-flange junction region of the I-profile, a region known to be prone to material imperfections [14].

Regarding the post-peak behaviour, the damage progression predicted by the models compares well with that observed experimentally, particularly considering the high experimental scatter and the fact that the models were developed using symmetry simplifications, and the asymmetric failure mode. More importantly, as illustrated in Fig. 27, the models were well able to capture the failure pattern development. The damage initiation occurred for very a very low displacement (0.08 mm, cf. Fig. 27a), due to crushing under the load application point and transverse tensile failure at that same section. Nevertheless, as discussed in Section 4.2, the “yield” transverse tensile stress was used as strength input (S_{22}^+), which is much lower than its ultimate strength (cf. Tables 1 and 2). Thereafter, in this case, the failure flag of the transversely tensioned elements depicted in Fig. 27 is not indicating that the elements fractured, but rather they have entered the damage progression stage (cf. Fig. 3). Afterwards, those initial damage modes progressed to surrounding elements leading to an interlaminar failure mode, due to a combination of through-thickness tensile and transverse interlaminar stresses (Fig. 27b). Such fracture progressed through the thickness direction (Fig. 27c) without reaching the load/support sections, as in the longitudinal interlaminar failure (cf. Fig. 25b), indicating that the trough-thickness tensile rupture contributed largely to this failure mode. Damage eventually reached the support sections (Fig. 27d), approximately when the maximum load was attained, via in-plane shear failure (as opposed to interlaminar shear failure), as observed in the experimental tests (Fig. 27e and f).

5. Conclusions

This paper presented a novel progressive failure model for the three-dimensional simulation of the mechanical behaviour of quasi-orthotropic pultruded FRP laminates with slender aspect ratios, which includes most commercial pultruded FRP profiles, using FE analysis and considering the material to be homogenous. The model is based on new failure initiation criteria, which takes into account the differences in the strength of FRP materials in various directions, and under compressive and tensile stresses. The damage progression is accounted for by (i) a modified MLT law, which considers the different failure evolution patterns in each direction, and (ii) a constant stress plateau after certain limit strains are reached.

The parameters of the model were calibrated based on a comprehensive experimental programme, which involved standard coupon tests performed on two pultruded GFRP plates. The results show that the proposed model, even considering the simplifying assumptions of through-thickness material homogeneity and uncoupled in-plane and

out-of-plane damages, is well able to simulate the failure modes and strengths obtained in the experiments, capturing very complex phenomena, such as interlaminar shear failure.

To further validate the proposed model and develop more comprehensive knowledge about the failure and post-failure behavior and mechanisms of pultruded FRP materials, compressive and tensile tests in the through-thickness are needed, a topic that should be addressed in future investigations. Additionally, the application of the proposed model to other pultruded materials, namely with different thicknesses and quasi-orthotropy ratios should also be object of future research.

A companion paper (Part II) will further attest the feasibility of the proposed damage progression model by applying it to two case studies, namely (i) compact tension tests, and (ii) web-crippling tests on full-section profiles.

6. Data availability

The raw/processed data required to reproduce these findings cannot be shared at this time as the data also forms part of an ongoing study. The UMAT containing the code of the proposed damage progression model will be made available after the paper is accepted for publication at the CORE Group website: <http://coregroup.tecnico.ulisboa.pt/~coregroup.daemon/publications/downloads/umat/>.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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